**Research Article**

# Induced Paths, Radius, and Extremal Neighbourhood Independence in Graphs without 4-Cycles

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## Abstract

Let $G$ be a connected graph on $n \geq 2$ vertices. Write $S(G)$ for the sum of the independence numbers of the open neighbourhoods, and write $\ell(G) = S(G)/n$. Every connected graph with no 4-cycle subgraph and no induced path on seven vertices satisfies $S \leq 3n$, with equality only for the Petersen graph. Combined with the classification of the connected graphs with no 4-cycle subgraph, radius 4 and induced-path order 7, this bound yields Conjecture 133 of *Written on the Wall II* for every connected graph with $\ell \leq 4$. The bound for graphs with at most one triangle is proved by hand; the bound for graphs with at least two triangles uses a finite search, cross-checked by a second implementation. The same arguments determine the maximum of $\ell$ over connected graphs with no 4-cycle subgraph and no induced path on six vertices, with a unique extremal graph for every order $n \geq 7$, and determine, for every $n \geq 10$, the maximum number of edges in a connected triangle-free graph with no 4-cycle subgraph and no induced path on seven vertices, together with all extremal graphs. The conjecture remains open for $\ell > 4$.

## 1 | Introduction

All graphs are finite and simple. For a graph $G$ with $n \geq 2$ vertices let $p(G)$ be the largest number of vertices of an induced path and let $r(G)$ be the radius. Write

$$S(G) = \sum_{v \in V(G)} \alpha\big(G[N(v)]\big), \qquad \ell(G) = \frac{S(G)}{n},$$

where $\alpha$ is the independence number and $N(v)$ the open neighbourhood. Thus $\ell(G)$ is the average, over the vertices, of the independence number of the open neighbourhood. Write $c_{C_4}(G) = 1$ if $G$ contains no 4-cycle as a subgraph and $c_{C_4}(G) = 0$ otherwise. A graph with no 4-cycle subgraph is called *strong $C_4$-free*. In the extremal literature the same condition is the usual meaning of $C_4$-free. In the induced-subgraph literature, including the theorems of Brandstädt and Hoàng, Karthick and Maffray, Chen and Xu, and Chen, Wu and Xu cited below, the same words mean that $C_4$ is forbidden only as an induced subgraph. The adjective *strong* records the subgraph condition. By contrast, $P_k$-free means that $G$ has no induced path on $k$ vertices.

A shortest path between two vertices at distance $\operatorname{diam}(G) \geq r(G)$ is induced, so $p(G) \geq r(G) + 1$. In a strong $C_4$-free graph any two vertices have at most one common neighbour, so each neighbourhood is a matching plus isolated vertices and $S(G) = 2e(G) - 3T(G)$.

**Conjecture 1.1** (Graffiti/WOWII Conjecture 133)**.** *Every connected graph $G$ on at least two vertices satisfies*

$$r(G) + \ell(G)^{c_{C_4}(G)} \le p(G).$$

The primary source is E. DeLaVina's list *Written on the Wall II*, produced by the conjecture-making program Graffiti.pc [6]. Entry 133, dated July 12, 2005, reads: if $G$ is a simple connected graph, then path$(G) \ge$ rad$(G)$ + [average of $l(v)$]$^{c\,C_4}$. On that list path$(G)$ is the largest number of vertices of an induced path and rad$(G)$ is the radius. Other entries write floors and ceilings in words, as FLOOR[$\cdots$] and CEIL[$\cdots$], so the brackets in entry 133 are grouping signs. The quantity $l(v)$ is the local independence of $v$, defined on the definitions page of the list as the independence number of the subgraph induced by the neighbours of $v$. The same page takes the exponent to be 1 when $G$ is $C_4$-free as a subgraph, not necessarily as an induced subgraph, and 0 otherwise. That is the convention in Conjecture 1.1. A record of the conjecture in the Formal Conjectures repository is a secondary source [8].

The real-valued reading, with no floor, is the stronger of the two readings compatible with the brackets. Remark 5.12 proves the floor form for every connected graph with $\ell < 5$. Since $p(G) - r(G)$ is an integer, the real-valued inequality on a strong $C_4$-free graph is equivalent to

$$p(G) - r(G) \;\ge\; \lceil \ell(G) \rceil. \tag{1.1}$$

If $G$ contains a 4-cycle, then $c_{C_4}(G) = 0$ and both readings reduce to $p(G) \ge r(G) + 1$.

The subgraph convention matters. Suppose the exponent were 1 precisely when $G$ has no induced 4-cycle. The book $K_2 \vee \overline{K_5}$ has order 7, radius 1 and induced-path order 3. Every 4-cycle of this book has a chord, and $\ell = 15/7$, so $r + \ell = 22/7 > 3$. The same count shows that $K_2 \vee \overline{K_m}$ fails the real-valued inequality for every $m \ge 5$. Under the subgraph convention each of these graphs has exponent 0, and the inequality is the trivial bound $p \ge r + 1$. The problem therefore reduces to strong $C_4$-free graphs.

The statements of independent interest are the sharp extremal theorems, not the partial case of the conjecture. Linear upper bounds on the number of edges, in classes with no long induced path and no $K_{s,s}$ subgraph, are classical. Galvin, Rival and Sands proved that a long path in a $K_{t,t}$-free graph forces a long induced path [9]. Kühn and Osthus proved that large average degree forces a $K_{s,s}$ subgraph or an induced subdivision of any fixed graph [12]. Loh, Tait, Timmons and Zhou introduced induced Turán numbers, in which one graph is forbidden as a subgraph and another only as an induced subgraph [13]. In that setting a linear bound with an identified extremal family is already a strong conclusion. The class here is smaller. It forbids $C_4$ as a subgraph, hence $K_{2,2}$, and it forbids a long induced path.

**Theorem 5.10.** Every connected strong $C_4$-free $P_7$-free graph $G$ satisfies $S(G) \le 3n$, with equality if and only if $G$ is the Petersen graph.

The triangle-free case is the following statement. For those graphs $S = 2e$.

**Theorem 5.5.** Let $G$ be a connected triangle-free strong $C_4$-free $P_7$-free graph. Then $2e(G) \le 3|V(G)|$, with equality if and only if $G$ is the Petersen graph.

The same class has an exact edge maximum. Write $f(n)$ for the maximum of $2e(G)$ over its members of order $n$.

**Theorem 5.6.** For every $n \ge 10$,

$$f(n) = 3n - \min\{n - 10,\ 5 + \mathbf{1}_{\{n \text{ even}\}}\}.$$

The extremal graphs are the graphs named in the statement of the theorem: the Petersen graph with $n - 10$ leaves on a single vertex or on the two ends of a single edge, together with the graphs $F_2$ and $F_0'$ and two one-leaf extensions of $F_2$, on the ranges of $n$ given there.

These three theorems are the headline. Conjecture 1.1 is the motivation for proving them and the application below. Two further structural statements are used in that application and have exact forms of their own.

**Theorem 3.6.** Let $M(n)$ be the maximum of $\ell$ over connected strong $C_4$-free $P_6$-free graphs of order $n \ge 7$. Then $M(7) = 16/7, M(8) = 5/2, M(9) = 8/3, M(10) = 3$ and $M(n) = 2 + 2/n$ for $n \ge 11$, and the extremal graph is unique for every such $n$. Every connected strong $C_4$-free $P_6$-free graph of order $n \ge 2$ satisfies $S \le 3n$, with equality only for the Petersen graph.

**Theorem 4.2.** A connected strong $C_4$-free graph of radius 4 and induced-path order 7 is $C_8$, or it is obtained from a 6-cycle by the extension in the statement of the theorem. If $b$ is the number of vertices added on edges of that 6-cycle and $m$ is the number of matching edges in the sets $C_i$, then $S = 2n - b - m$. In particular $S \le 2n$.

**Theorem 5.11.** Every connected graph on at least two vertices with $\ell \le 4$ satisfies Conjecture 1.1.

The proof of Theorem 5.10 splits on the number of triangles. Theorem 5.5 is the triangle-free case. Its input is the classification, due to Chen and Xu, of the connected $(P_7, C_4, C_3)$-free graphs with no clique cutset, the observation that a 2-connected member of the triangle-free class has no clique cutset, and a decomposition of a general member into such a core with trees attached at single vertices. Comparing the deficits $3|V(Q)| - 2e(Q)$ of the cores gives Theorem 5.6 as well. Theorem 5.7 is the case of at most one triangle: a minimal counterexample is 2-connected, and deleting a minimum-degree vertex of the triangle returns the argument to a triangle-free core. If there are at least two triangles, Lemma 5.8 lists the induced subgraphs formed by two triangles, and Lemma 5.9 produces a vertex of deletion loss at most 3 by a finite search. The search has no order cutoff and no isomorphism pruning. At each node it selects the vertex of deletion loss less than 4 with the fewest legal children, breaking ties by the smallest label, and it explores every legal child. The completeness argument is in the proof of Lemma 5.9. A node of a search is unrestricted when every vertex of the current graph has deletion loss at least 4. Each of the four search trees is finite and has no unrestricted node, so the branch that follows a counterexample ends at a leaf. At that leaf the chosen vertex has no legal new neighbour, while the counterexample supplies one. Appendix A prints the decision arrays. A second implementation of the same search, supplied with a JGT referee report (2026), is archived in `referee_checks/`.

Theorem 5.11 then assembles the radius cases. It uses Lemmas 3.1, 3.2, 3.3 and 3.5. Through Lemma 3.5 it also uses Lemmas 3.4 and 2.8, the Camby–Schaudt theorem and Seinsche's theorem. It uses the Erdős–Saks–Sós theorem; Theorem 3.6, in the case $r = 2$ and again when $r = 3$ and the induced-path order is at most 5; Corollary 4.4 through Theorem 4.2, together with the observation that $r = 4$ and $p \ge 8$ give $p \ge r + \ell$ whenever $\ell \le 4$; and Theorem 5.10. Theorem 5.10 uses Theorems 5.5 and 5.7, Lemmas 5.8 and 5.9, and the Chen–Xu classification. Lemma 5.9 enters only through Theorem 5.10, and only for graphs with at least two triangles. In the proof of Theorem 5.11 that use occurs only for $r = 3$, $3 < \ell \le 4$ and $p = 6$. If $3 < \ell \le 4$ and $p \le 5$, Theorem 3.6 gives $\ell \le 3$. If $p = 6$, Theorem 5.10 gives $\ell \le 3$.

Since $\lambda(v) \le d(v)$ for every vertex, $\ell \le 2e/n$. Theorem 5.11 covers every connected graph of average degree at most 4. For a strong $C_4$-free graph the counting bound of Reiman gives $\ell \le (1+\sqrt{4n-3})/2$, as recorded after the proof of Theorem 5.11. The right-hand side is at most 4 whenever $n \le 13$. Theorem 5.11 covers every connected graph on at most 13 vertices: a graph that contains a 4-cycle has exponent 0, and a strong $C_4$-free graph on at most 13 vertices has $\ell \le 4$. The same method gives $p \ge r + \ell$ whenever $\ell \le r - 1$, because the Erdős–Saks–Sós theorem supplies an induced path on at least $2r - 1$ vertices. It does not reach $\ell > 4$. The obstruction is the absence of an analogue of Theorem 5.10 for longer induced paths, for instance a bound $S \le 4n$ for connected strong $C_4$-free $P_9$-free graphs. Section 6 reduces any counterexample with $\ell > 4$ to a failure of $\mathrm{GEN}_K$ for some $K \ge 4$ or of $\mathrm{RAD}(r)$ for some $r \ge 5$, and it records the open conjecture on $S$ together with four examples.

There is a parallel structure theory in which $C_4$ is forbidden only as an induced subgraph. Brandstädt and Hoàng describe $(P_6, C_4)$-free graphs by clique cutsets [1]. Karthick and Maffray describe the square-free members of that class [10]. Chen and Xu classify the connected $(P_7, C_4, C_3)$-free graphs with no clique cutset [5], and that classification is the core list for Theorems 5.5 and 5.6. Chen, Wu and Xu describe further $(P_7, C_4)$-free classes, with $C_4$ induced [4]. Those theorems do not by themselves yield Theorems 3.6, 5.5 or 5.6. The strong $C_4$-free class excludes every graph with a 4-cycle subgraph, including a nontrivial clique blow-up of the Petersen graph and a join of the Petersen graph with a nonempty clique, and it contains graphs with clique cutsets. The exact value of $M(n)$ uses the second-layer count on an induced 6-cycle, which needs the subgraph condition. Lemma 2.9 is the emptiness list on an induced 5-cycle, proved directly in Section 2 from $P_6$-freeness, induced-$C_4$-freeness and the absence of an induced $C_6$. The colouring bound of Karthick and Maffray is a different statement.

In [3], Chen studies the floor form $p \ge r + \lfloor \ell \rfloor^{c_{C_4}}$ of the same conjecture. That deposit is not a refereed paper, so every statement taken from it and used below is proved again in this article. Three points separate the two texts.

1. In [3], a graph with no 4-cycle subgraph satisfies $\ell = (2e - 3T)/n$. On an induced 6-cycle in a $P_6$-free graph of that class, an outside vertex has empty trace or meets the cycle in an antipodal pair. Under an auxiliary hypothesis RES, namely radius 2, diameter 3 and $\ell > 3$, the second layer is restricted, and five admissible frames of orders 6 through 9, all with $\ell < 3$, are listed. The diameter-2 theorem in that deposit is likewise stated under $\ell > 3$.
2. The present article determines $M(n)$ for every order, with a unique extremal graph for each $n \ge 7$; classifies the radius-4 graphs of induced-path order 7; gives $S \le 3n$ for every connected strong $C_4$-free $P_7$-free graph, with equality only for the Petersen graph; determines $f(n)$ and all extremal graphs for every $n \ge 10$; and proves the real-valued conjecture for $\ell \le 4$, with no diameter hypothesis and no appeal to RES.

3. The identity $S = 2e - 3T$ is proved as Lemma 2.1, and the antipodal-trace constraint is proved, with the second-layer rule, in the proof of Lemma 3.4 and in Case 1 of Theorem 3.6. Those arguments stand on their own, so no step of this article depends on the unrefereed deposit [3].

Section 3 proves Theorem 3.6. Section 4 proves Theorem 4.2. Section 5 proves Theorems 5.5, 5.6, 5.7 and 5.10, and Theorem 5.11. Section 6 is the concluding section. It reduces the range $\ell > 4$ to two open statements and records the open conjecture on $S$ together with four examples. The search of Lemma 5.9 is recorded in Appendix A.

## 2 | Preliminaries

For $v \in V(G)$ let $d(v)$ be its degree and $t(v)$ the number of triangles containing $v$. Write $\lambda(v) = \lambda_G(v) = \alpha(G[N(v)])$, $S(G) = \sum_v \lambda(v)$, $e(G)$ for the number of edges and $T(G)$ for the number of triangles.

**Lemma 2.1.** *Let $G$ be strong $C_4$-free. Then:*

1. *two distinct vertices have at most one common neighbour;*
2. *for every $v$, $G[N(v)]$ is a matching together with isolated vertices, so $\lambda(v)$ equals the number of components of $G[N(v)]$ and $\lambda(v) = d(v) - t(v)$;*
3. *$S(G) = 2e(G) - 3T(G)$.*

*Proof.* (1) Two common neighbours $w, w'$ of $x, y$ give the 4-cycle $xwyw'$. (2) A vertex of $N(v)$ with two neighbours in $N(v)$ would have two common neighbours with $v$. So the components are single vertices and edges, each edge being a triangle at $v$. (3) Sum $d(v) - t(v)$ over $v$. □

The components of $G[N(v)]$ are the *units* of $v$. For a vertex set $U$, $G - U$ is the subgraph induced by $V \setminus U$, and $D_G(U) = S(G) - S(G - U)$ is the *deletion loss* of $U$. We write $L(v) = D_G(\{v\})$.

**Lemma 2.2.** *Let $G$ be strong $C_4$-free and $\emptyset \neq U \subseteq V(G)$. Then*

$$D_G(U) = \sum_{u \in U} \lambda(u) + \sum_{x \notin U} \#\{\text{units of } x \text{ contained in } U\}.$$

*In particular $L(v) = 2\lambda(v) - t(v) = 2d(v) - 3t(v)$.*

*Proof.* For $x \notin U$, deleting $U$ removes exactly the units of $x$ that lie inside $U$; a unit that is an edge with one end in $U$ survives as a vertex. For $U = \{v\}$: the $2t(v)$ neighbours of $v$ in triangles with $v$ keep their value, and each of the other $d(v) - 2t(v)$ neighbours loses one. The total is $\lambda(v) + d(v) - 2t(v) = 2\lambda(v) - t(v)$. □

**Lemma 2.3.** *Let $v$ be a cut vertex of $G$. Split the components of $G - v$ into two nonempty groups, and let $G_1, G_2$ be the subgraphs induced by the two groups together with $v$. Then $S(G) = S(G_1) + S(G_2)$.*

*Proof.* $G[N(v)]$ has no edges between the groups, so $\lambda_G(v) = \lambda_{G_1}(v) + \lambda_{G_2}(v)$. Every other vertex keeps its neighbourhood. □

We use three published results.

**Theorem 2.4** (Erdős, Saks and Sós [7])**.** *Every connected graph $G$ has an induced path on at least $2r(G) - 1$ vertices.*

**Theorem 2.5** (Camby and Schaudt [2])**.** *Let $k \geq 4$ and let $G$ be a connected graph with no induced path on $k$ vertices. Then $G$ has a connected dominating set $X$ such that $G[X]$ has no induced path on $k-2$ vertices or $G[X]$ is isomorphic to the cycle $C_k$.*

**Theorem 2.6** (Seinsche [15])**.** *If a graph on at least two vertices has no induced path on four vertices, then it or its complement is disconnected.*

Theorem 2.5 is implication (i)⇒(ii) of Theorem 5 of arXiv:1402.7213, applied with $H = G$. In that preprint, Theorem 5 is the equivalence, for every connected induced subgraph, between $P_k$-freeness and the existence of such a connected dominating set. The journal article [2] states the same equivalence. We write $P_k$ for the path on $k$ vertices, and "$G$ is $P_k$-free" means that $G$ has no induced $P_k$, that is, $p(G) \leq k - 1$. A connected graph of radius $r$ has at least $2r$ vertices.

**Lemma 2.7.** *Let $H$ be a connected strong $C_4$-free $P_4$-free graph on at least two vertices. Then $H$ has a vertex adjacent to all other vertices.*

*Proof.* By Theorem 2.6, $H$ is the join of two nonempty vertex sets $A$ and $B$. If $|A|, |B| \geq 2$, then $a, a' \in A$ and $b, b' \in B$ span the 4-cycle $aba'b'$. So one side is a single vertex, and it is adjacent to everything. □

**Lemma 2.8.** *Let $G$ be a connected graph with no induced path on six vertices. Then $G$ has a connected dominating set $X$ such that $G[X]$ has no induced path on four vertices, or else $G[X]$ is isomorphic to $C_6$.*

This is Theorem 2.5 at $k = 6$. Lemma 3.5 uses the resulting dominating set.

**Lemma 2.9** (Karthick and Maffray [10, Theorem 3.2(d)])**.** *Let $G$ be a graph with no induced $P_6$, no induced $C_4$ and no induced $C_6$, and let $C = c_0c_1c_2c_3c_4c_0$ be an induced 5-cycle, indices modulo 5. For each $i$ let $W_i$ be the set of vertices outside $C$ whose neighbourhood on $C$ is exactly $\{c_i\}$, and let $X_{i,i+1}$ be the set of vertices outside $C$ whose neighbourhood on $C$ is exactly $\{c_i, c_{i+1}\}$. Then, for every $i$, at least one of $X_{i,i+1}$ and $X_{i+1,i+2}$ is empty, at least one of $W_i$ and $W_{i+2}$ is empty, and at least one of $X_{i,i+1}$ and $W_{i-1} \cup W_{i+2}$ is empty.*

*Proof.* The three assertions are Theorem 3.2(d) in Karthick and Maffray. The argument here is self-contained. Fix $i$ and write indices modulo 5. Throughout, $C$ is induced, a vertex of $W_i$ meets $C$ only at $c_i$, and a vertex of $X_{i,i+1}$ meets $C$ only at $c_i$ and $c_{i+1}$.
Let $x \in X_{i,i+1}$ and $y \in X_{i+1,i+2}$. If $x \sim y$, the vertices $x, c_i, c_{i-1}, c_{i+3}, c_{i+2}, y$ induce a 6-cycle. The vertex $c_{i+1}$ lies off this set, so the edge $c_ic_{i+1}$ is not a chord, $x$ has no further neighbour among the six vertices except $c_i$ and $y$, and $y$ has no further neighbour among them except $c_{i+2}$ and $x$. If $x \nsim y$, the path $y, c_{i+2}, c_{i+3}, c_{i-1}, c_i, x$ is an induced $P_6$: again only one cycle-neighbour of $x$ lies on the path.
Let $x \in W_i$ and $y \in W_{i+2}$. If $x \sim y$, the vertices $x, c_i, c_{i-1}, c_{i+3}, c_{i+2}, y$ induce a 6-cycle. If $x \nsim y$, the same six vertices, in that order and without the edge $xy$, induce a $P_6$.
Let $x \in X_{i,i+1}$ and $y \in W_{i-1}$. If $x \sim y$, the vertices $y, c_{i-1}, c_{i+3}, c_{i+2}, c_{i+1}, x$ induce a 6-cycle. If $x \nsim y$, the same vertices in that order induce a $P_6$. If instead $y \in W_{i+2}$ and $x \sim y$, then $x, c_{i+1}, c_{i+2}, y$ induce a 4-cycle, since $x \nsim c_{i+2}$ and $y \nsim c_{i+1}$. If $x \nsim y$, the path $y, c_{i+2}, c_{i+3}, c_{i-1}, c_i, x$ is an induced $P_6$. □

A strong $C_4$-free graph has no induced $C_4$. Whenever it is $P_6$-free and the case of an induced 6-cycle has been excluded, it meets the hypotheses of the lemma. Case 2 of Theorem 3.6 uses the first two assertions. The third assertion is part of the list in [10] and is not required there.

**Lemma 2.10.** *Let $G$ be a connected triangle-free strong $C_4$-free $P_7$-free graph with no cut vertex. Then $G$ has no clique cutset.*

*Proof.* The graph is triangle-free, so every clique has order at most two, and there is no clique cutset of order one. Suppose $\{u, v\}$ is a clique cutset. Then $uv$ is an edge. Neither vertex is a cut vertex, so every component of $G - \{u, v\}$ is adjacent to both $u$ and $v$, and there are at least two such components. A shortest $u$–$v$ path through one component has at least four edges: one internal vertex would form a triangle with $uv$, and two internal vertices would form a 4-cycle with $uv$. On each of two such paths, let $x, y, z$ be the three vertices immediately preceding $v$, in that order from the $u$-side, and write $x', y', z'$ for the same choice in the second component. A shortest $u$–$v$ path has at least four edges, so $x, y, z$ and $x', y', z'$ are internal vertices and in particular distinct from $u$. They therefore lie in two distinct components of $G - \{u, v\}$, and there is no edge between the two triples. Each triple is a subpath of a shortest path, so it has no chord. The path $x, y, z, v, z', y', x'$ is an induced $P_7$, which contradicts $p(G) \le 6$. □

## 3 | Graphs of induced-path order at most five

**Lemma 3.1.** *If $G$ is connected and strong $C_4$-free with $r(G) = 1$, then $S(G) = 2(n-1) - m$, where $m$ is the number of edges of $G[N(u)]$ for a dominating vertex $u$. Hence $\ell(G) < 2$, and (1.1) holds.*

*Proof.* Let $v \neq u$. By Lemma 2.1(1), $v$ has at most one neighbour other than $u$, and such a neighbour lies in $N(u)$. So $G[N(v)]$ is a clique and $\lambda(v) = 1$; also $\lambda(u) = (n-1) - m$. Hence $S = 2(n-1) - m < 2n$ and $\lceil \ell \rceil \le 2$. If $G$ is complete then $G \in \{K_2, K_3\}$, $\ell = 1$ and $p = 2 = r + 1$. Otherwise two nonadjacent vertices of $N(u)$ together with $u$ form an induced $P_3$, so $p \ge 3 \ge r + \lceil \ell \rceil$. □

**Lemma 3.2.** *If $G$ is connected and strong $C_4$-free with $r(G) \ge 2$, then $p(G) \ge 4$.*

*Proof.* If $G$ were $P_4$-free, Lemma 2.7 would give a dominating vertex, so $r(G) \le 1$. □

**Lemma 3.3.** *If $G$ is connected, strong $C_4$-free and $P_5$-free, then $S(G) \le 2n - h$, where $h$ is the number of vertices $v$ with $\lambda(v) \ge 3$. In particular $S(G) \le 2n$.*

*Proof.* Call $v$ *simplicial* if $\lambda(v) = 1$ and *high* if $\lambda(v) \ge 3$. For $x \in N(v)$ put $X_x = N(x) \setminus N[v]$, and let $P_x$ be the set of neighbours of $x$ in $G[N(v)]$. The set $P_x$ is empty when $x$ is isolated in $G[N(v)]$, and otherwise it consists of the other end of the unique edge unit containing $x$. Thus $N(x) \subseteq \{v\} \cup P_x \cup X_x$.

*Key fact.* Let $x, y$ lie in different units of $v$. A common vertex of $X_x$ and $X_y$ would give the 4-cycle $vxwy$, so $X_x \cap X_y = \emptyset$. If $x' \in X_x$ and $y' \in X_y$ are nonadjacent, then $x'xvyy'$ is an induced $P_5$: $x \nsim y$ because they lie in different units, $x \nsim y'$ and $y \nsim x'$ by disjointness, and $v$ is adjacent to neither $x'$ nor $y'$. So $X_x$ is complete to $X_y$. If both are nonempty and $|X_x| \geq 2$, a vertex of $X_y$ has two neighbours in $X_x$, which are also neighbours of $x$, a contradiction to Lemma 2.1(1). Hence if both are nonempty, then $|X_x| = |X_y| = 1$.
*Main step.* For each high $v$ we find $\lambda(v) - 1$ simplicial vertices whose only high neighbour is $v$, one in each of $\lambda(v) - 1$ distinct units of $v$.
Case 1: some $u \in N(v)$ has $|X_u| \geq 2$. By the key fact $X_y = \emptyset$ for every $y$ outside the unit of $u$. Then $N(y) \subseteq \{v\} \cup P_y$. This set is a clique, so $y$ is simplicial, and so is the other vertex of its unit when that unit is an edge. Hence $v$ is the only high neighbour of $y$, and the $\lambda(v) - 1$ units other than that of $u$ supply the vertices.
Case 2: $|X_x| \leq 1$ for all $x \in N(v)$. Call a unit *active* if one of its vertices $x$ has $X_x \neq \emptyset$. Suppose two units are active, with $y' \in X_y$ and $z' \in X_z$, and let $x$ lie in a third unit. By the key fact $y' \sim z'$, and $xvyy'z'$ is an induced $P_5$: $x \nsim y$ (different units), $x \nsim y', z'$ (disjointness), $v \nsim y', z'$, and $y \nsim z'$ (disjointness). As $\lambda(v) \geq 3$, a third unit exists, so at most one unit is active. Every vertex of an inactive unit is simplicial and its only high neighbour is $v$. This gives at least $\lambda(v) - 1$ units.
*Counting.* The chosen vertices for distinct high vertices are distinct, since each has a unique high neighbour. So $\sum_{v \text{ high}}(\lambda(v) - 1) \leq \#\{\text{simplicial vertices}\}$, and

$$S = \sum_{v} \lambda(v) \leq 2n - \#\{\text{simplicial}\} + \sum_{v \text{ high}} (\lambda(v) - 2) \leq 2n - h. \qquad \square$$

**Lemma 3.4.** *Let $G$ be connected, strong $C_4$-free and $P_6$-free with a dominating induced cycle $C = v_0v_1 \cdots v_5$, and let $W = V(G) \setminus V(C)$. Then every $w \in W$ is adjacent to exactly one antipodal pair $\{v_i, v_{i+3}\}$ and to no other vertex of $C$, distinct vertices of $W$ use distinct pairs, and $W$ is independent. Hence $|W| \leq 3$, $S(G) = 12 + 4|W| < 3n$, and $r(G) \leq 2$ if $W \neq \emptyset$.*

*Proof.* Indices are modulo 6. Let $w \in W$; it has a neighbour on $C$. If $w$ has exactly one neighbour $v_i$, then $wv_iv_{i+1}v_{i+2}v_{i+3}v_{i+4}$ is an induced $P_6$. If $N(w) \cap C = \{v_i, v_{i+1}\}$, then $wv_{i+1}v_{i+2}v_{i+3}v_{i+4}v_{i+5}$ is an induced $P_6$. Two neighbours at distance 2 on $C$ give a 4-cycle. A set of at least three vertices of $C_6$ contains two vertices at distance 2. So $N(w) \cap C$ is an antipodal pair. Two vertices of $W$ with the same pair give two common neighbours. If $w \sim w'$ with distinct pairs, some $v_i \in N(w)$ and $v_j \in N(w')$ are adjacent on $C$, and $wv_iv_jw'$ is a 4-cycle. There are no triangles, $e(G) = 6 + 2|W|$, and Lemma 2.1(3) gives $S = 12 + 4|W|$. Finally, if $w$ uses $\{v_i, v_{i+3}\}$, then every vertex of $C$ is within distance 2 of $v_i$ (the vertex $v_{i+3}$ through $w$), and each $w' \in W$ has a neighbour in $\{v_{i-1}, v_i, v_{i+1}\}$, because each of the three antipodal pairs meets this set. So $v_i$ has eccentricity 2. $\square$

The same trace analysis applies to any vertex adjacent to an induced 6-cycle in a connected strong $C_4$-free $P_6$-free graph, whether or not the cycle dominates. Theorem 3.6 uses that form. Karthick and Maffray [10, Theorem 3.3] classify the external neighbourhood of an induced $C_6$ in a $(P_6, C_4)$-free graph. When the graph has no 4-cycle subgraph, only the empty trace and the antipodal-pair traces remain. The new content here is the second-layer constraint, the resulting order bound, the exact value of $M(n)$, and uniqueness.

**Lemma 3.5.** *Let $G$ be connected, strong $C_4$-free and $P_6$-free. Then either $G$ has a vertex of eccentricity at most 2, or $G = C_6$.*

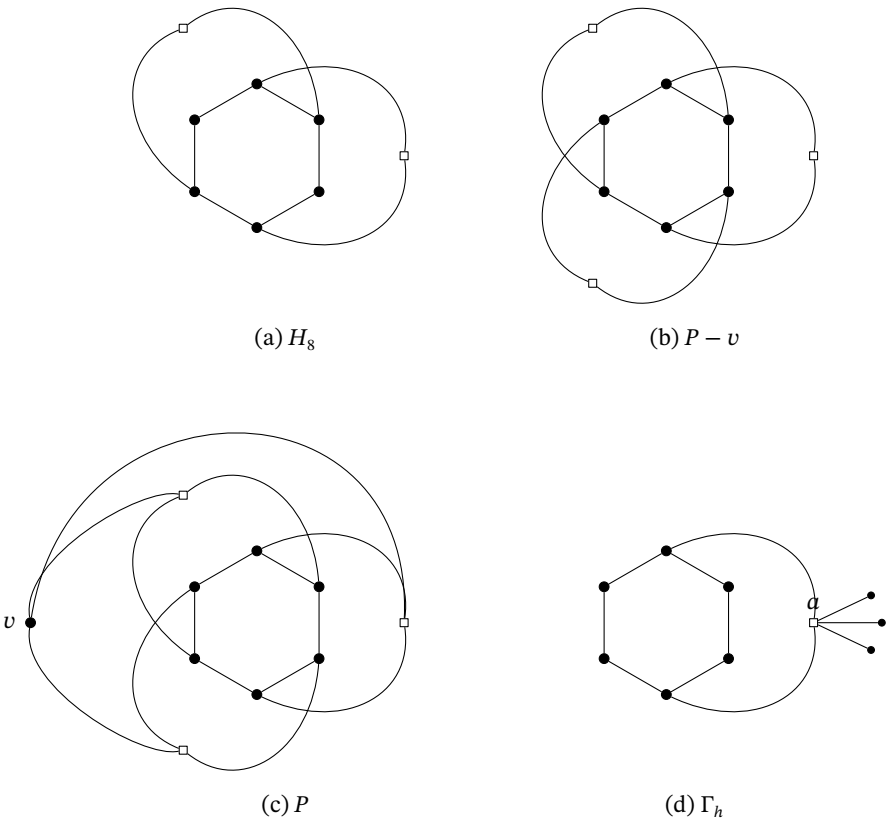
*Proof.* Lemma 2.8 supplies a connected dominating set $X$. If $G[X] \cong C_6$, Lemma 3.4 gives $G = C_6$ or a vertex of eccentricity 2. Otherwise $G[X]$ is $P_4$-free. If $|X| = 1$, the vertex in $X$ dominates $G$. If $|X| \geq 2$, Lemma 2.7 gives $u \in X$ adjacent to all of $X \setminus \{u\}$, and since $X$ dominates $G$, every vertex is within distance 2 of $u$. $\square$

**Theorem 3.6.** *Let $M(n)$ be the maximum of $\ell(G)$ over connected strong $C_4$-free $P_6$-free graphs of order $n \geq 7$. Then*

$$M(n) = \begin{cases} 16/7 & \text{if } n = 7, \\ 5/2 & \text{if } n = 8, \\ 8/3 & \text{if } n = 9, \\ 3 & \text{if } n = 10, \\ 2 + 2/n & \text{if } n \geq 11. \end{cases}$$

*For every $n \geq 7$ the extremal graph is unique up to isomorphism. For $n = 10$ it is the Petersen graph $P$. For $n = 9$ it is $P - v$. For $n = 8$ it is the graph $H_8$ of Figure 1. For $n = 7$ and for every $n \geq 11$ it is the graph $\Gamma_{n-7}$ of that figure: a 6-cycle, a vertex $a$ adjacent exactly to one antipodal pair, and $n - 7$ leaves adjacent only to $a$. For $2 \leq n \leq 6$ the same maximum $M(n)$, and the extremal graphs, are as follows: $n = 2$, $M(2) = 1$, only $K_2$; $n = 3$, $M(3) = 4/3$, only $P_3$; $n = 4$, $M(4) = 3/2$, exactly $P_4$ and $K_{1,3}$; $n = 5$, $M(5) = 2$, only $C_5$; $n = 6$, $M(6) = 2$, exactly $C_6$ and the graph obtained from $C_5$ by adding one pendant*

*leaf. Since $M(n) \le 3$ for every $n \ge 2$, every connected strong $C_4$-free $P_6$-free graph $G$ of order $n \ge 2$ satisfies $S(G) \le 3n$, with equality if and only if $G$ is the Petersen graph.*




(a) $H_8$ (b) $P - v$

(c) $P$ (d) $\Gamma_h$

**FIGURE 1** | Extremal graphs for $M(n)$. The six central filled vertices form the cycle, and each open square meets one antipodal pair. (a) $H_8$, order 8, $\ell = 5/2$. (b) $P - v$, order 9, $\ell = 8/3$. (c) The Petersen graph, order 10, $\ell = 3$. These three graphs are triangle-free, so $S = 2e$. (d) $\Gamma_h$: any number of leaves at the square $a$, including none. The case $h = 0$ is the unique graph of order 7 with $\ell = 16/7$. For every $n \ge 11$ the extremal graph is $\Gamma_{n-7}$, with $S = 2n+2$.

*Proof.* Throughout, $S = 2e - 3T$ by Lemma 2.1. A chordless cycle of length at least seven contains an induced $P_6$. Thus the only possible holes have lengths five and six.

*Case 1: an induced six-cycle.* Label its vertices $c_0, \dots, c_5$, indices modulo 6, and write $A$ for the set of vertices at distance one from the cycle. The trace argument in the proof of Lemma 3.4 shows that each $a \in A$ meets the cycle in one antipodal pair. Distinct vertices of $A$ use distinct pairs, and $A$ is an independent set: an edge of $A$ joins distinct pairs, and some contact of one end is adjacent on the cycle to some contact of the other, giving a 4-cycle. Hence $k = |A| \le 3$. Figure 1 shows the extremal graphs of this branch in panels (a)–(c), for $k \in \{2, 3\}$, and the family $\Gamma_h$ in panel (d), for $k = 1$. Figure 2 shows the two distance layers used in the count.

Every vertex $x$ at distance two from the cycle is adjacent to every vertex of $A$. Such an $x$ has at least one neighbour in $A$. If $x \sim a$ and $x \nsim b$ for $a, b \in A$, choose contacts so that $c_i \in N(a)$ and $c_{i+2} \in N(b)$. This is possible: the pairs are distinct, so some contact of $b$ lies at cycle distance two from some contact of $a$, and we orient the cycle from that contact of $a$ toward that contact of $b$. The path $x, a, c_i, c_{i+1}, c_{i+2}, b$ is then an induced $P_6$. Indeed $x$ meets the cycle in nothing, $a$ meets the cycle only in its antipodal pair (neither $c_{i+1}$ nor $c_{i+2}$), $b$ meets the cycle only in its antipodal pair (so $b \sim c_{i+2}$ and $b \nsim c_i, c_{i+1}$), and $a \nsim b$.

There is no vertex at distance three or more from the cycle. If there were, the cycle would have a vertex $z$ at distance exactly three. A shortest $z$–cycle path is then the path $z, x, a, c_i$ on four vertices, with $x$ at distance two and $a$ at distance one. Extending that path by $c_{i+1}, c_{i+2}$ gives an induced $P_6$: the vertex $z$ has no neighbour in $A$ or on the cycle, and $x$ has no neighbour on the cycle.

Let $h$ be the number of vertices at distance two. If $k = 0$, then $A$ is empty, so the cycle has no neighbour and $G = C_6$. If $k = 1$, every distance-two vertex is adjacent to the unique vertex $a \in A$. The distance-two vertices induce a matching together

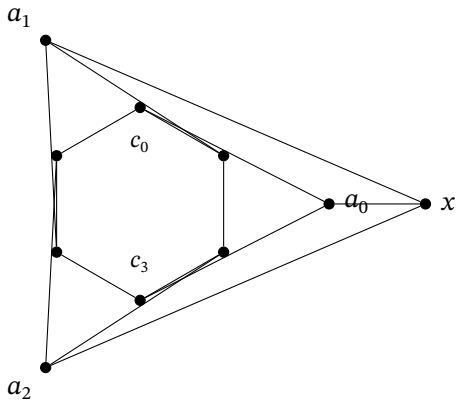



**FIGURE 2** | Distance layers of an induced 6-cycle in Case 1 of Theorem 3.6. Each $a_i$ meets one antipodal pair, the pairs are distinct, and the first layer is independent, so $k \le 3$. A vertex $x$ of the second layer meets every vertex of the first layer and none of the cycle. If $k \in \{2, 3\}$ that layer has size at most one, and the order is at most 10. There is no third layer.

with isolated vertices: a vertex of that layer with two neighbours in the layer would have two common neighbours with $a$. Writing $m$ for the number of edges of the matching gives $n = 7+h, e = 8+h+m$ and $T = m$, so $S = 16+2h-m = 2n+2-m$. Thus $S \le 2n + 2$, with equality if and only if $m = 0$. Equality means that $G$ is the graph $\Gamma_h$ formed by the 6-cycle, the vertex $a$ on one antipodal pair, and $h$ leaves at $a$.

The graph $\Gamma_h$ lies in the class, for every $h \ge 0$. A leaf has degree one. Every other cycle through an outside vertex uses both edges from $a$ to the cycle, and each of the two $a$-paths along the cycle has length three, so every such cycle has length five. There is no 4-cycle. There is no induced $P_6$. A leaf, having degree one, occupies an end of any path that contains it, and that path meets $a$ next. Two leaves therefore determine a path on only three vertices. An induced path that contains one leaf and five further vertices would restrict to an induced path of order five in $\Gamma_0$ that starts at $a$. No such path exists: the two neighbours of $a$ on the cycle are an antipodal pair, the path $a, c_i, c_{i+1}, c_{i+2}, c_{i+3}$ has the chord $ac_{i+3}$, and every shorter continuation from $a$ through one neighbour stops at the vertex before the antipode, because the next cycle vertex is adjacent to $a$. An induced path of $\Gamma_0$ that avoids $a$ lies on the 6-cycle and has order at most five. An induced path of $\Gamma_0$ that contains $a$ in its interior has $c_i$ and $c_{i+3}$ as the two neighbours of $a$ on the path. Each side can then take at most one further cycle vertex: the next vertex on either arc is adjacent to the antipode already used. The resulting path has order at most five. Thus $\Gamma_h$ is strong $C_4$-free and $P_6$-free, and it attains $S = 2n + 2$.

If $k \in \{2, 3\}$, then $h \le 1$: two vertices of the second layer would both be adjacent to two vertices of $A$. There are no triangles. The second-layer vertex, if it exists, is adjacent to every vertex of $A$ and to nothing on the cycle, $A$ is independent, and each vertex of $A$ contributes two edges to the cycle. Thus $n = 6 + k + h$ and $S = 2(6 + 2k + kh)$. The four possibilities are

| $(k, h)$ | $(2, 0)$ | $(2, 1)$ | $(3, 0)$ | $(3, 1)$ |
|---|---|---|---|---|
| $S/n$ | $5/2$ | $8/3$ | $8/3$ | $3$ |

The four pairs are the graphs of Figure 1, with $(3, 0)$ and $(2, 1)$ both equal to $P - v$. Write $a_i$ for the vertex adjacent to the antipodal pair $\{c_i, c_{i+3}\}$. The pair $(k, h) = (3, 1)$ is the graph in panel (c): the three vertices $a_0, a_1, a_2$ and a vertex $v$ adjacent to all three. It is cubic of order 10. It is triangle-free, and the trace rules give no 4-cycle, so its girth is at least 5. Its diameter is 2. Indeed every pair of cycle vertices is at distance at most 2 along the cycle or through the corresponding $a_i$; the vertex $v$ meets every $a_i$; and a non-neighbour of $a_i$ on the cycle is adjacent to $c_i$ or to $c_{i+3}$. Suppose $v_0 \cdots v_5$ were an induced path. Diameter 2 gives $v_0$ a common neighbour with each of $v_3, v_4, v_5$. Those neighbours lie off the path, because the only neighbour of $v_0$ on the path is $v_1$ and the path is induced. They are distinct: a vertex adjacent to two of $v_3, v_4, v_5$ would form a triangle with a path edge, or a 4-cycle with the length-2 subpath $v_3v_4v_5$. This is impossible: $v_0$ has degree 3, so it has only two neighbours off the path. Thus the graph is $P_6$-free. It is the Petersen graph $P$, and $\ell(P) = 3$.

Deleting $v$ leaves $(k, h) = (3, 0)$, the graph $P - v$ in panel (b). The pair $(2, 1)$ is isomorphic to the same graph. In the labelling with $A = \{a_0, a_1\}$ and second-layer vertex $v$, the cycle $c_0, a_0, c_3, c_4, a_1, c_1$ has its three antipodal pairs served by $c_5$, $v$ and $c_2$, and no vertex remains. Both graphs therefore have $\ell = 8/3$, and both are the graph $P - v$. Induced subgraphs of $P$ are strong $C_4$-free and $P_6$-free. The pair $(2, 0)$ is the graph $H_8$ in panel (a), with $\ell = 5/2$. The three choices of the unused antipodal pair differ by a rotation, so this graph is unique up to isomorphism. Its order is 8, so it is neither $P$ nor $P - v$. For $k \in \{2, 3\}$ one has $n = 6 + k + h \le 10$.

*Case 2: an induced five-cycle and no induced six-cycle.* Write $C = c_0 \cdots c_4 c_0$. Every vertex at distance one from $C$ has a singleton or consecutive-pair trace: any other pair of contacts lies at distance two on $C$ and forms a 4-cycle. If a vertex $x$ were at distance two, with a neighbour $a$ in the first layer, the path $x, a$ followed by four consecutive cycle vertices, beginning at one contact of $a$ and omitting the second contact if $a$ has one, would be an induced $P_6$. Thus $C$ dominates $G$. Write $h$ for the number of vertices with singleton trace and $j$ for the number with a consecutive-pair trace. At most one vertex carries each of the five pair traces, since two would have two common neighbours, so $j \leq 5$. The edges outside $C$ are of the following kinds.

- Two singleton vertices with different supports are nonadjacent. Neighbouring supports give a 4-cycle. Supports at distance two give an induced 6-cycle along the longer arc of $C$. Edges within one singleton class form a matching, because two neighbours in that class would be two common neighbours of the supporting cycle vertex. Let $m$ be the number of these edges.
- No two pair vertices are adjacent. From their two distinct support edges one can choose adjacent cycle vertices, one from each edge, and those two vertices together with the pair vertices form a 4-cycle.
- A singleton supported at $c_i$ meets only a pair vertex supported on the opposite edge $\{c_{i+2}, c_{i+3}\}$. Every other pair edge contains a neighbour of $c_i$ and yields a 4-cycle. That pair vertex meets at most one vertex of the singleton class, again by the common-neighbour bound. Let $b$ be the number of singleton-to-pair edges, so $b \leq \min(h, j)$.

Consequently $n = 5 + h + j$ and $e = 5 + h + 2j + m + b$. The triangles are the $j$ pair triangles and the $m$ matching triangles, so $T = j + m$ and

$$S = 2e - 3T = 2n + 2b - j - m.$$

The integer $b$ is at most 1. Suppose two singleton–pair edges are present. They use two distinct singleton supports. The opposite edge of $c_i$ is $\{c_{i+2}, c_{i+3}\}$. As $i$ varies, these five edges exhaust $C$, and each pair class contains at most one vertex. The graph is $P_6$-free and strong $C_4$-free, and the present case assumes that it has no induced $C_6$, so Lemma 2.9 applies. Under that lemma the set $W_i$ is the singleton class at $c_i$, and the set $X_{i,i+1}$ is the consecutive-pair class on the edge $\{c_i, c_{i+1}\}$. If the two singleton supports lie at distance 2 on $C$, they are $W_i$ and $W_{i+2}$ for some $i$, and both are nonempty. Lemma 2.9 states that at least one of those two sets is empty. If the supports lie at distance 1, take them to be $c_i$ and $c_{i+1}$. The opposite edge of $c_i$ is $\{c_{i+2}, c_{i+3}\}$ and the opposite edge of $c_{i+1}$ is $\{c_{i+3}, c_{i+4}\}$. Both pair classes are nonempty, one for each of the two edges, and they are $X_{i+2,i+3}$ and $X_{i+3,i+4}$. Lemma 2.9 states that at least one of $X_{j,j+1}$ and $X_{j+1,j+2}$ is empty for every $j$; the index $j = i + 2$ is this pair of classes. Either alternative contradicts the existence of both edges, so $b \leq 1$.
If $b = 0$, then $S = 2n - j - m \leq 2n$. If $b = 1$, then $j \geq 1$ because $b \leq \min(h, j)$, so $S = 2n + 2 - j - m \leq 2n + 1$. In this case $S \leq 2n + 1$.
*Case 3: no holes.* The graph is chordal, and every nonempty induced subgraph has a simplicial vertex. The neighbourhood of a simplicial vertex is a clique. The degree is at most two: four vertices of a clique contain a 4-cycle. A simplicial vertex of a connected chordal graph on at least two vertices is not a cut vertex, and the graph left by deleting it is chordal and strong $C_4$-free. In the graph present at the deletion, the loss is $L(v) = 2d(v) - 3t(v)$. Degree one gives loss 2. Degree two, with the two neighbours adjacent, gives one triangle and loss 1. Deleting vertices until a single vertex remains, one has $S \leq 2(n - 1) = 2n - 2$.
The three cases give the values of $M(n)$. A graph of order $n \geq 11$ cannot realise $k \in \{2, 3\}$ in Case 1, and $k = 0$ forces $G = C_6$. The only remaining graphs in Case 1 are the graphs $\Gamma_h$ and the graphs with the same first layer and $m \geq 1$. The latter have $S \leq 2n + 1$. Cases 2 and 3 give $S \leq 2n + 1$ and $S \leq 2n - 2$. Thus $M(n) = 2 + 2/n$, attained only by $\Gamma_{n-7}$.
The graphs with $S > 2n + 2$ are as follows. At order $n = 10$ the only graph in the list is $P$, with $S = 30$. At order $n = 9$ the only such graph is $P - v$, with $S = 24$, whether it arises as $(k, h) = (3, 0)$ or as $(k, h) = (2, 1)$. At order $n = 8$ the only such graph is $H_8$, with $S = 20$. At order $n = 7$ every graph in the list satisfies $S \leq 2n + 2$. One has $h = 0$ in the equality case of $k = 1$, so the unique graph with $S = 2n + 2$ is $\Gamma_0$, while Cases 2 and 3 give at most 15 and 12. Since $8/3 < 3$ and $2 + 2/n < 3$ for $n \geq 11$, the bound $S \leq 3n$ holds for every order $n \geq 2$, with equality only for $P$.
The case division does not use $n \geq 7$. At order 2 the only connected graph is $K_2$, with $S = 2$ and $\ell = 1$. At order 3 there is no cycle of length five or six, so only the chordal bound applies and $S \leq 4$. Equality holds for the tree $P_3$, while $K_3$ has $S = 3$. At order 4 the same bound gives $S \leq 6$, with equality for a tree, and the only trees on four vertices are $P_4$ and $K_{1,3}$. At order 5 the induced 6-cycle branch is empty. In the induced 5-cycle branch one has $h = j = 0$, so $G = C_5$ and $S = 10$. The chordal branch contributes at most 8. At order 6 the induced 6-cycle branch contributes only $C_6$, with $S = 12$. In the induced 5-cycle branch one has $h + j = 1$, hence $b = 0$ and $S = 12 - j - m \leq 12$, with equality precisely when $j = m = 0$ and $h = 1$, the graph $C_5$ with one leaf. The chordal branch contributes at most 10. Thus the maximum is 12, attained by exactly those two graphs. □

## 4 | Radius four

**Lemma 4.1.** *Let $G$ be connected, strong $C_4$-free and $P_8$-free. If an induced 8-cycle dominates $G$ and $r(G) \geq 4$, then $G = C_8$.*

*Proof.* Write $C = c_0c_1 \cdots c_7c_0$, with indices modulo 8. For an outside vertex $y$ the trace $A_y = N(y) \cap V(C)$ is nonempty. The trace $A_y$ is not contained in any three consecutive vertices of $C$. Suppose $A_y \subseteq \{c_i, c_{i+1}, c_{i+2}\}$. The ends $c_i$ and $c_{i+2}$ are not both in $A_y$, since $y, c_i, c_{i+1}, c_{i+2}$ would be a 4-cycle. Thus $A_y$ is a singleton or a consecutive pair. A singleton $\{c_i\}$ gives the induced path

$$y, c_i, c_{i+1}, c_{i+2}, c_{i+3}, c_{i+4}, c_{i+5}, c_{i+6}.$$

A consecutive pair $\{c_i, c_{i+1}\}$ gives the induced path

$$y, c_i, c_{i-1}, c_{i-2}, c_{i-3}, c_{i-4}, c_{i-5}, c_{i-6},$$

in which the omitted contact $c_{i+1}$ does not lie on the path. Either path is an induced $P_8$.
Fix a cycle vertex, say $c_0$. The vertices at cycle distance greater than two from $c_0$ are the three consecutive vertices $c_3, c_4, c_5$. The trace of any outside vertex meets the complementary set $\{c_7, c_0, c_1, c_2, c_6\}$, so every outside vertex is within distance three of $c_0$. The same holds with $c_0$ replaced by any vertex of $C$.
Suppose $w$ lies outside $C$, and write $A = N(w) \cap V(C)$. No two vertices of $A$ are at cycle distance two, for they would form a 4-cycle with $w$. A set of vertices of $C$ in which every two distinct vertices are consecutive has size at most two, and every set of size at most two that is a singleton or a consecutive pair is contained in a triple of consecutive vertices. Both possibilities have been excluded. Therefore $A$ contains two vertices at cycle distance three or four. Rotating, take them to be $c_0$ and $c_d$ with $d \in \{3, 4\}$.
The antipode of $c_0$ is $c_4$. If $d = 4$, the path $c_0wc_4$ has length two. If $d = 3$, then $c_3 \sim c_4$, so $c_0wc_3c_4$ has length three. Every other vertex of $C$ is already at cycle distance at most three from $c_0$. Every outside vertex is within distance three of $c_0$. Thus the eccentricity of $c_0$ is at most three, contradicting $r(G) \geq 4$.
Hence $C$ has no vertex outside it, and $G = C_8$. □

**Theorem 4.2.** *A connected graph $G$ is strong $C_4$-free and satisfies $r(G) = 4$ and $p(G) = 7$ if and only if either $G \cong C_8$, or $G$ is obtained from a 6-cycle $C = v_0v_1 \cdots v_5$ by the following construction and by no further vertices or edges. For each $i$ there is a nonempty set $C_i$ of vertices outside $C$. Every vertex of $C_i$ is adjacent to $v_i$ and to no other vertex of $C$. There is no edge between $C_i$ and $C_j$ for $i \neq j$, and $G[C_i]$ is a matching together with isolated vertices. For each edge of $C$ there is at most one further vertex, adjacent precisely to the two ends of that edge.*

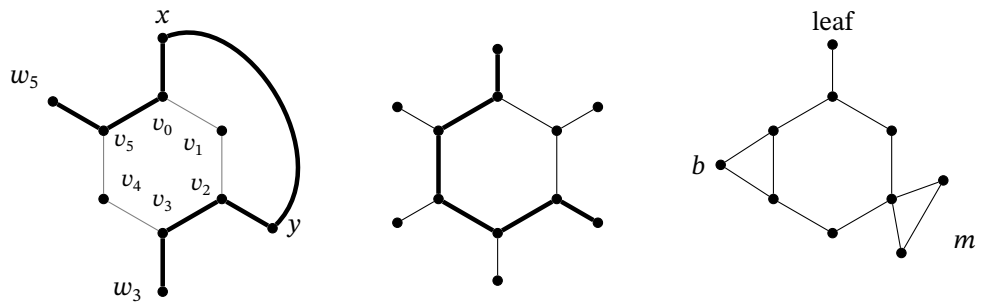



**FIGURE 3** | The radius-four case of Theorem 4.2. (a) The induced $P_8$ is $w_5, v_5, v_0, x, y, v_2, v_3, w_3$, with singleton traces $\{v_0\}$ and $\{v_2\}$ at $x$ and $y$, and with $w_5$, $w_3$ the distance-four witnesses. (b) One leaf at each cycle vertex gives $S = 2n$. (c) Only local attachments are shown; the remaining nonempty attachment classes are omitted. An isolated leaf does not contribute to $b + m$. A matching edge in some $C_i$, and a vertex glued to a cycle edge, each increase $2n - S = b + m$ by 1.

**Lemma 4.3.** *Let $G$ be obtained from a 6-cycle $C$ by the construction of Theorem 4.2. Let $P$ be an induced path in $G$ of order at least three. Every vertex of $P$ that does not lie on $C$ is an endpoint of $P$. In particular $P$ contains at most two vertices outside $C$, and any such vertex is an endpoint of $P$.*

*Proof.* Let $z$ be a vertex of $P$ outside $C$. If $z$ is an isolated vertex of $G[C_i]$ for some $i$, then $z$ has degree one, so $z$ is an endpoint of every path that contains it.

If $z$ lies on a matching edge of $G[C_i]$, the neighbourhood of $z$ is $\{v_i, z'\}$, where $z'$ is the other end of that edge, and this neighbourhood is a clique. Suppose $z$ were internal on $P$. An internal vertex has two neighbours on $P$. Those neighbours are then $v_i$ and $z'$, which are adjacent, and the path uses the route $v_i z z'$ or $z' z v_i$. The edge $v_i z'$ joins the two neighbours of $z$ and skips $z$, so it is a chord of $P$. An induced path has no chord. Thus $z$ is an endpoint of $P$. The same chord shows that $P$ does not contain $z$, $z'$ and $v_i$ together.
If $z$ was added on a cycle edge, the neighbourhood of $z$ is that edge, again a clique. The same chord argument shows that $z$ is an endpoint of $P$, and that $P$ does not contain both ends of the cycle edge together with $z$.
A path has two endpoints. Every vertex of $P$ outside $C$ is one of them, so there are at most two such vertices. □

*Proof of Theorem 4.2.* First suppose that $G$ is strong $C_4$-free, with $r(G) = 4$ and $p(G) = 7$. Apply Theorem 2.5 with $k = 8$. The hypotheses hold: $k \geq 4$, and $G$ is connected and has no induced path on eight vertices because $p(G) = 7$. The conclusion is a connected dominating set $X$ such that $G[X]$ is $P_6$-free or $G[X] \cong C_8$.
If $G[X] \cong C_8$, Lemma 4.1 gives $G = C_8$.
If $G[X]$ is $P_6$-free, apply Lemma 3.5 inside $G[X]$. A vertex of eccentricity at most two in $G[X]$ has eccentricity at most three in $G$, because $X$ dominates $G$, contradicting $r(G) = 4$. Therefore $G[X] = C_6$. Let $C = v_0 \cdots v_5$ be this dominating induced 6-cycle, indices modulo 6, and let $W = V(G) \setminus V(C)$.
*Traces.* As in the proof of Lemma 3.4, strong $C_4$-freeness allows only three kinds of trace $N(w) \cap C$: a single vertex, a consecutive pair $\{v_i, v_{i+1}\}$, or an antipodal pair. If some $w$ has trace $\{v_i, v_{i+3}\}$, then $v_i$ is within distance two of every vertex of $C$, hence within distance three of every vertex of $G$, a contradiction. So there are no antipodal traces. Let $C_i$ be the set of vertices with trace $\{v_i\}$.
*Witnesses.* Fix $i$. Since $r(G) = 4$, some vertex is at distance at least four from $v_{i+3}$. Every vertex of $C$ is within distance three of $v_{i+3}$, and a vertex $w \in W$ is at distance at most $1 + \min\{d_C(v_{i+3}, x) : x \in N(w) \cap C\}$ from $v_{i+3}$, which is at most three unless the trace of $w$ is $\{v_i\}$. So there is $w_i \in C_i$ at distance four from $v_{i+3}$. Every neighbour $z \in W$ of $w_i$ lies in $C_i$: if $z$ had a neighbour among $v_{i+2}, v_{i+3}, v_{i+4}$, then $d(w_i, v_{i+3}) \leq 3$; if $z$ had the neighbour $v_{i-1}$ or $v_{i+1}$, then $w_i, v_i, v_{i\pm1}, z$ would be a 4-cycle. So $N(w_i) \subseteq \{v_i\} \cup C_i$.
*Edges in $W$.* Figure 3 records the two configurations used below. Let $x, y \in W$ be adjacent.

- $x \in C_0$, $y \in C_{\pm1}$: $x, v_0, v_{\pm1}, y$ is a 4-cycle.
- $x \in C_0$, $y \in C_2$: the path $w_5, v_5, v_0, x, y, v_2, v_3, w_3$ of Figure 3(a) is an induced $P_8$. The 6-cycle is induced, so it contributes no chord among $v_5, v_0, v_2, v_3$. Each off-cycle vertex meets the cycle only at its designated vertex: $w_5$ at $v_5$, $x$ at $v_0$, $y$ at $v_2$ and $w_3$ at $v_3$. The witness neighbourhoods, already proved, satisfy $N(w_5) \subseteq \{v_5\} \cup C_5$ and $N(w_3) \subseteq \{v_3\} \cup C_3$, so neither witness is adjacent to $x$, to $y$, or to the other witness. These three facts leave no chord.
- $x \in C_0$, $y \in C_3$: $w_1, v_1, v_0, x, y, v_3, v_4, w_4$ is an induced $P_8$. By symmetry, vertices of $C_i$ and $C_j$ with $i \neq j$ are never adjacent.
- $x$ with trace $\{v_0, v_1\}$ and $y \in C_j$: for $j \in \{0, 1\}$ the vertices $v_0, v_1, x, y$ span a 4-cycle; for $j = 5$ (respectively $j = 2$) the cycle $x, v_0, v_5, y$ (respectively $x, v_1, v_2, y$) is a 4-cycle; for $j = 3$, $w_5, v_5, v_0, x, y, v_3, v_2, w_2$ is an induced $P_8$, and $j = 4$ is symmetric.
- $x, y$ with consecutive-pair traces $\{v_0, v_1\}$ and $\{v_j, v_{j+1}\}$: $j = 0$ gives two common neighbours; $j = \pm1$ and $j = \pm2$ give 4-cycles as above; $j = 3$ gives the induced $P_8$ $w_2, v_2, v_1, x, y, v_4, v_5, w_5$.

The remaining paths displayed above are induced for the same three reasons, applied to the witnesses and traces named in those bullets. So every edge inside $W$ joins two vertices of the same class $C_i$, and these edges form a matching. A consecutive-pair vertex is adjacent to no vertex of any $C_j$ and to no other consecutive-pair vertex. Two vertices with the same consecutive-pair trace would have two common neighbours, so each cycle edge carries at most one such vertex, and that vertex is adjacent precisely to the two ends of the edge. Each class $C_i$ is nonempty. This is the stated construction.
Conversely, let $G$ be $C_8$, or let $G$ be built from $C = v_0 \cdots v_5$ as in the statement. The cycle $C_8$ has radius 4, induced-path order 7 and no 4-cycle subgraph. Now consider the construction on $C$. There is no edge between distinct sets $C_i$, and every vertex of $C_i$ meets $C$ only at $v_i$. A cycle that meets $C_i$ is therefore contained in $G[C_i \cup \{v_i\}]$. In that subgraph the edges are a star at $v_i$ together with a matching, so the only cycles are the triangles formed by $v_i$ and one matching edge.
Let $y$ be a vertex added on the edge $\{v_j, v_{j+1}\}$, so $N(y) = \{v_j, v_{j+1}\}$. Every cycle through $y$ uses both edges at $y$, and the rest of the cycle is a $v_j$–$v_{j+1}$ path avoiding $y$. A 4-cycle through $y$ is then a path of length two between $v_j$ and $v_{j+1}$ that avoids $y$: the two ends of the edge have a common neighbour other than $y$. No vertex of any $C_i$ is adjacent to two vertices of $C$, and the construction places at most one vertex on each cycle edge. Those two ends therefore have no common neighbour other than the vertex added on that edge. Hence no 4-cycle passes through $y$. The only cycle that uses no vertex outside $C$ is $C$ itself. Thus $G$ has no 4-cycle subgraph.

Let $P$ be an induced path of order at least three. By Lemma 4.3, every vertex of $P$ outside $C$ is an endpoint of $P$, so every other vertex of $P$ lies on $C$. The cycle $C$ has no induced path on six vertices. Hence $p(G) \leq 7$. Since each $C_i$ is nonempty, one may choose $x \in C_0$ and $y \in C_2$. The path $x, v_0, v_5, v_4, v_3, v_2, y$ is induced: $x$ and $y$ meet the path only at $v_0$ and $v_2$, and $C$ contributes no chord. Thus $p(G) = 7$.

There is no edge between distinct sets $C_i$, and a vertex added on a cycle edge has degree two on that edge, so it does not shorten distances along $C$. The distance from $v_i$ to $v_{i+3}$ is therefore three. A vertex of $C_{i+3}$ is at distance four from $v_i$, and the eccentricity of $v_i$ is at least four. It is at most four: every vertex of $C_j$ is at distance $d_C(v_i, v_j) + 1 \leq 4$ from $v_i$, and every vertex added on a cycle edge is within distance four of $v_i$. A vertex of $C_i$ is at distance five from a vertex of $C_{i+3}$. A vertex $y$ added on the edge $\{v_j, v_{j+1}\}$ has only those two neighbours. The distances from $v_j$ and from $v_{j+1}$ to $v_{j+3}$ are 3 and 2, neither neighbour of $y$ is adjacent to $v_{j+3}$, and $y, v_{j+1}, v_{j+2}, v_{j+3}$ is a path of length three, so a vertex of $C_{j+3}$ is at distance four from $y$. Every eccentricity is at least four, and $r(G) = 4$. □

**Corollary 4.4.** *Let $G$ be connected and strong $C_4$-free with $r(G) = 4$ and $p(G) = 7$. In the construction of Theorem 4.2, let $b$ be the number of vertices added on edges of the 6-cycle and let $m$ be the number of matching edges in the sets $C_i$. Then $S(G) = 2n - b - m$. In particular $S(G) \leq 2n$, so $\ell(G) \leq 2$.*

*Proof.* For $G \cong C_8$ one has $n = 8, e = 8, T = 0$ and $S = 16 = 2n$, which is the formula with $b = m = 0$. In the construction, write $a = \sum_i |C_i|$. Then $n = 6 + a + b$. The edges are the six cycle edges, $a$ edges from the sets $C_i$ to the cycle, $m$ matching edges and $2b$ edges incident with the vertices added on cycle edges, so $e = 6 + a + 2b + m$. Each matching edge forms one triangle with its cycle vertex, and each added vertex forms one triangle with its cycle edge, so $T = b + m$. Lemma 2.1(3) gives $S = 2e - 3T = 2n - b - m$. □

**Corollary 4.5.** *Let $G$ be connected and strong $C_4$-free with $r(G) = 4$ and $p(G) = 7$. Then $S(G) = 2n$ if and only if $G \cong C_8$, or $G$ is obtained from $C_6$ by attaching at least one leaf to each cycle vertex.*

*Proof.* By Corollary 4.4, $S = 2n$ if and only if $G \cong C_8$ or, in the construction, $b = m = 0$. In the latter case every vertex outside the cycle is a leaf, and each $C_i$ is nonempty, so each cycle vertex has at least one leaf. The converse is the case $b = m = 0$ of the construction, together with $C_8$, both of which have $S = 2n$ and satisfy the numerical hypotheses by Theorem 4.2. □

## 5 | Graphs of induced-path order at most six

In a triangle-free graph, a 4-cycle subgraph is an induced 4-cycle, and every clique has order at most two. The next classification is Lemma 3.4 of Chen and Xu [5]. Their family $\mathfrak{F} = \{F_0, F_0', F_1, \ldots, F_{12}\}$ is defined by the constructions in the proofs of their Lemmas 3.1–3.3; $F_5$ is the Petersen graph, and the parameters $t$ and $t'$ attached to $F_2$ and $F_0'$ are positive integers. The edge counts below are taken from those constructions, as reproduced in the list that follows.

**Lemma 5.1** (Chen and Xu [5, Lemma 3.4])**.** *Let $G$ be a connected $(P_7, C_4, C_3)$-free graph with no clique cutset. If $G$ is not complete, then $G \in \mathfrak{F}$.*

The members of $\mathfrak{F}$ admit the following descriptions, taken from those proofs.

- $F_1$ is $C_5$. The graph $F_2$ is obtained from a 5-cycle $v_1 \cdots v_5$ by adding stable sets $A_1$ and $A_3$ of equal positive order $t$, with a perfect matching between them, each vertex of $A_1$ adjacent only to $v_1$ and its match, and each vertex of $A_3$ adjacent only to $v_3$ and its match.
- On the same 5-cycle, $F_3$ has three further vertices inducing the path $a_3a_1a_4$, with $a_i$ adjacent to $v_i$. The graph $F_4$ has four further vertices inducing the path $a_5a_3a_1a_4$, with the same spokes. The graph $F_5$ is the Petersen graph.
- $F_0$ is $C_6$. The graph $F_0'$ is obtained from a 6-cycle $v_1 \cdots v_6$ by adding stable sets $Y_1$ and $Y_4$ of equal positive order $t'$, with a perfect matching between them, each vertex of $Y_1$ adjacent only to $v_1$ and its match, and each vertex of $Y_4$ adjacent only to $v_4$ and its match.
- $F_6$ is $C_7$. On a 7-cycle $v_1 \cdots v_7$, write $x_i$ for a vertex with neighbourhood $\{v_i, v_{i+3}\}$ on the cycle. Then $F_7$ is the cycle plus $x_1$; $F_8$ is the cycle plus nonadjacent $x_1, x_2$; $F_9$ is $F_8$ plus a vertex adjacent only to $x_1$ and $x_2$; $F_{10}$ is the cycle plus nonadjacent $x_1, x_5$; $F_{11}$ is the cycle plus a stable set $\{x_1, x_2, x_5\}$; and $F_{12}$ is $F_{11}$ plus a vertex adjacent only to $x_1$ and $x_2$.

Chen and Xu record that $F_0, F_1$, the graph $F_2$ with $t = 1$, $F_3$ and $F_4$ are induced subgraphs of the Petersen graph, and that $F_6, \ldots, F_{11}$ are induced subgraphs of $F_{12}$. The graphs $F_6, \ldots, F_{12}$ are the extensions of $C_7$ by vertices of cycle-distance three. The graphs $F_2$ and $F_0'$ are the matching extensions of one pair of opposite vertices on $C_5$ and on $C_6$.

**Lemma 5.2.** *Every graph in $\mathfrak{F}$ other than the Petersen graph satisfies $2e \le 3n-3$. The Petersen graph satisfies $2e = 3n$.*

*Proof.* The constructions give the following orders and sizes. For $F_2$ and $F_0'$ the parameters $t, t'$ are at least one.

| Graph | $n$ | $e$ | $2e$ | $3n-2e$ |
|---|---|---|---|---|
| $F_1 = C_5$ | 5 | 5 | 10 | 5 |
| $F_2$ | $5+2t$ | $5+3t$ | $10+6t$ | 5 |
| $F_3$ | 8 | 10 | 20 | 4 |
| $F_4$ | 9 | 12 | 24 | 3 |
| $F_5$ | 10 | 15 | 30 | 0 |
| $F_0 = C_6$ | 6 | 6 | 12 | 6 |
| $F_0'$ | $6+2t'$ | $6+3t'$ | $12+6t'$ | 6 |
| $F_6 = C_7$ | 7 | 7 | 14 | 7 |
| $F_7$ | 8 | 9 | 18 | 6 |
| $F_8, F_{10}$ | 9 | 11 | 22 | 5 |
| $F_9, F_{11}$ | 10 | 13 | 26 | 4 |
| $F_{12}$ | 11 | 15 | 30 | 3 |

The five edges of $F_3$ beyond $C_5$ are the three spokes and the two edges of $a_3a_1a_4$. The seven edges of $F_4$ beyond $C_5$ are the four spokes and the three edges of $a_5a_3a_1a_4$. Each vertex $x_i$ contributes two edges, and each vertex placed on a pair $x_ix_j$ contributes two further edges. The Petersen graph is the only row with difference zero. □

**Lemma 5.3.** *Let $G$ be connected and strong $C_4$-free, and let $s$ be a vertex at which no induced $P_4$ starts. Write $m = e(G[N(s)])$. Then $S(G) \le 2(|G|-1) - m$.*

*Proof.* A vertex at distance 3 from $s$ would end a shortest path, hence an induced $P_4$, starting at $s$. So $V = \{s\} \cup N_1 \cup N_2$ with $N_i$ the $i$-th distance layer. Each $y \in N_2$ has a unique parent $a \in N_1$. A neighbour $z \in N_2$ of $y$ with a different parent would give the induced path $sayz$. So $y$ has at most one neighbour in $N_2$, a sibling (two siblings would be common neighbours of $a$ and $y$), and $N(y)$ is a clique; thus $\lambda(y) = 1$. For $a \in N_1$ we have $N(a) \subseteq \{s, \bar{a}\} \cup C_a$, where $C_a$ is the set of children of $a$ and $\bar{a}$ is the neighbour of $a$ in $N_1$, if any. Since $s$ and $\bar{a}$ lie in one unit, $\lambda(a) \le 1 + |C_a|$. Finally $\lambda(s) = |N_1| - m$. Summing, $S \le (|N_1| - m) + (|N_1| + |N_2|) + |N_2| = 2(|G|-1) - m$. □

**Lemma 5.4.** *Let $G$ be a connected triangle-free strong $C_4$-free $P_7$-free graph of order $n$. Then $G$ is obtained from a connected triangle-free strong $C_4$-free $P_7$-free graph $Q$ with no clique cutset by attaching trees at single vertices. The core $Q$ is $K_1$, $K_2$, or a graph in the list of Lemma 5.2. Writing $h = |V(Q)|$ and $\eta(Q) = 3h - 2e(Q)$, one has $e(G) = e(Q) + (n-h)$ and*

$$3n - 2e(G) = \eta(Q) + n - h.$$

*Deleting the vertices of one attached tree, other than its root on $Q$, deletes exactly as many edges as vertices.*

*Proof.* If $G$ itself has no cut vertex, Lemma 2.10 gives that $G$ has no clique cutset. The hypotheses of that lemma are the present hypotheses together with the absence of a cut vertex. Lemma 5.1 then applies. Its hypotheses are that $G$ is connected, that $G$ has no induced $P_7$, that $G$ has no induced $C_4$ and no triangle, and that $G$ has no clique cutset. Connectedness and the absence of a clique cutset hold by the case and by Lemma 2.10. The graph is $P_7$-free by hypothesis. A strong $C_4$-free graph has no induced $C_4$, and a triangle-free graph has no $C_3$. If $G$ is not complete, Lemma 5.1 places it in the family $\mathfrak{F}$, hence on the list of Lemma 5.2. The only complete triangle-free graphs are $K_1$ and $K_2$. In every subcase one may take $Q = G$. The identities are then immediate. The core $K_1$ is the whole graph only when $n = 1$.
Suppose $s$ is a cut vertex. Two induced paths of order four that start at $s$ and otherwise lie in distinct components of $G - s$ concatenate to an induced $P_7$. Thus some component $C$ of $G - s$ contains no induced $P_4$ starting at $s$, and every induced path of $G[C \cup \{s\}]$ that starts at $s$ has order at most three. The first distance layer of this side is independent, since an edge there would form a triangle with $s$. There is no vertex at distance three or more from $s$, for such a vertex would end an induced $P_4$ at $s$. A vertex of the second layer has a unique parent: a second parent, together with $s$, would span a 4-cycle. It has no neighbour in its layer. An edge joining vertices of distinct parents is the induced path of order four recorded in the proof of Lemma 5.3, and an edge joining two children of the same parent forms a triangle with that parent. Thus $G[C \cup \{s\}]$ is a tree. Deleting the $k = |C|$ vertices of the side deletes exactly $k$ edges. The graph that remains is connected, triangle-free, strong $C_4$-free and $P_7$-free. Iterate. The remaining graph $Q$ has no cut vertex, and the first paragraph describes it. Reversing the deletions attaches trees at single vertices. Each deleted vertex accounted for one edge, so $e(G) = e(Q) + (n-h)$, and the displayed identity follows. The same deletion shows that $K_1$ is the core of every tree in the class, of any order. A connected $P_7$-free tree is triangle-free and strong $C_4$-free, and deleting leaves until one vertex remains leaves that core. □

**Theorem 5.5.** *Let $G$ be a connected triangle-free strong $C_4$-free $P_7$-free graph. Then $2e(G) \le 3|V(G)|$, with equality if and only if $G$ is the Petersen graph.*

*Proof.* Lemma 5.4 gives a core $Q$ of order $h$ and the identity

$$3n - 2e(G) = \eta(Q) + n - h.$$

Lemma 5.2 gives $\eta(Q) \geq 0$ for every graph on its list, with equality only for the Petersen graph. The cores $K_1$ and $K_2$ have deficits 3 and 4. The right-hand side is therefore nonnegative, and it vanishes if and only if $\eta(Q) = 0$ and $n = h$. Thus equality holds if and only if $G$ is the Petersen graph. □

Lemma 5.1 classifies the connected graphs in the class with no clique cutset. The extremal count is obtained by attaching trees at single vertices of such a core and comparing the resulting deficits, for every order $n \geq 10$. Figure 4 draws the families that meet the minimum.

**Theorem 5.6.** *Let $f(n)$ be the maximum of $2e(G)$ over connected triangle-free strong $C_4$-free $P_7$-free graphs of order $n$. For every $n \geq 10$,*

$$f(n) = 3n - \min\{n - 10,\ 5 + \mathbf{1}_{\{n \text{ even}\}}\}.$$

*Write $\mathcal{P}(n)$ for the isomorphism classes of graphs obtained from the Petersen graph by placing $n - 10$ leaves on a single vertex or on the two endpoints of a single edge. For $10 \leq n \leq 14$ the extremal graphs are exactly the classes in $\mathcal{P}(n)$. For $n = 15$ they are the classes in $\mathcal{P}(15)$ together with $F_2$, four classes in all. For $n = 16$ they are the classes in $\mathcal{P}(16)$, the graph $F_0'$, and the two graphs obtained from the order-15 graph $F_2$ by adding a leaf at a hub or at the midpoint of its unique 2-edge branch, seven classes in all. For odd $n \geq 17$ the unique extremal graph is $F_2$. For even $n \geq 18$ there are exactly three: $F_0'$, and the two graphs obtained from the order-$(n-1)$ graph $F_2$ by adding a leaf at a hub or at the midpoint of its unique 2-edge branch.*

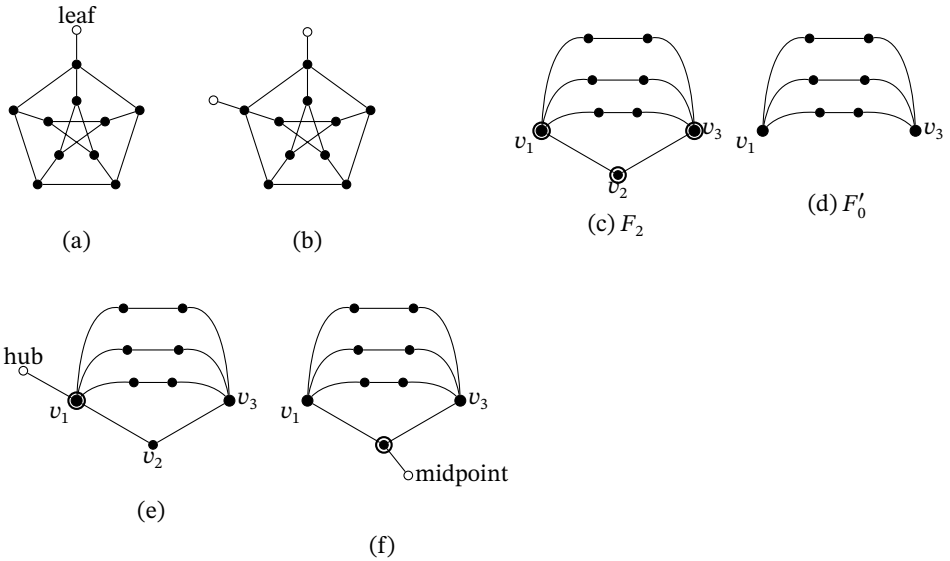



**FIGURE 4** | Equality families of Theorem 5.6. (a) and (b) Leaves of $\mathcal{P}(n)$ at one vertex, or at both ends of one edge. Both have deficit $n - 10$. (c) $F_2$: length-3 paths on distinct arcs, and one length-2 path through $v_2$. Deficit 5. Rings mark both hubs and the midpoint. (d) $F_0'$: every hub-to-hub path has length 3, on its own arc. Deficit 6. (e) and (f) The allowed leaf sites, a hub and the short-branch midpoint. The isomorphism classes at orders 15 and 16 are listed in Table 1.

*Proof.* Write $\eta(H) = 3|V(H)| - 2e(H)$ for a triangle-free graph $H$.
*Reduction to one core.* Lemma 5.4 supplies a core $Q$ of order $h$ and the identity $3n - 2e(G) = \eta(Q) + n - h$.
*Upper bound.* Every fixed row of the table in Lemma 5.2 satisfies $\eta \geq h - 10$. Each such graph has order at most 10 except $F_{12}$, which has order 11 and $\eta = 3$, and $3 \geq 11 - 10$. The same inequality holds for $K_1$ and $K_2$, with deficits 3 and 4. For every such core, $\eta(Q) + n - h \geq n - 10$. The two parameterised rows are $F_2$, of odd order and deficit 5, and $F_0'$, of even order and deficit 6. If the core is $F_2$ and the final order is even, then at least one vertex has been attached, so the displayed deficit is at least 6. If the core is $F_2$ and the final order is odd, the deficit is at least 5. If the core is $F_0'$, the deficit is at least 6. In every case

$$3n - 2e(G) \geq \min\{n - 10,\ 5 + \mathbf{1}_{\{n \text{ even}\}}\}.$$

*Attainment.* The value $f(n)$ is attained by the graphs named in the statement; the next paragraph shows that no other graph attains it. A leaf added to a triangle-free strong $C_4$-free graph creates neither a triangle nor a cycle. Realise the Petersen graph on the 2-element subsets of $\{1,2,3,4,5\}$, with two subsets adjacent when they are disjoint. This is the graph $P$ of Theorem 3.6. The symmetric group on the ground set acts on the vertices. The action is vertex-transitive and edge-transitive, and it carries any two nonadjacent vertices to $\{1,2\}$ and $\{1,3\}$. The path

$$\{1,2\},\{3,4\},\{1,5\},\{2,3\},\{1,4\}$$

is an induced $P_5$ starting at $\{1,2\}$, and

$$\{1,2\},\{3,4\},\{1,5\},\{2,4\},\{1,3\}$$

is an induced $P_5$ with ends $\{1,2\}$ and $\{1,3\}$. Thus every vertex of $P$ is an end of an induced $P_5$, and every pair of nonadjacent vertices is the pair of ends of an induced $P_5$.

Place $a$ leaves at one end of an edge of $P$ and $b$ leaves at the other, with $a+b=n-10$ and $a \geq b \geq 0$. The case $b=0$ is the placement at a single vertex. The graph is connected, triangle-free and strong $C_4$-free. It has $15+(n-10)$ edges, so its deficit is $n-10$. An induced path uses a leaf only as an end. A path that uses two leaves at the same vertex has order three. A path that uses one leaf at each end of the chosen edge is the edge itself with the two leaves attached: the ends of that edge are adjacent, so no longer induced path of $P$ has those ends. A path that uses one leaf restricts to an induced path of $P$ and has order at most six, because $P$ is $P_6$-free. Thus the graph is $P_7$-free. Vertex-transitivity and edge-transitivity give one isomorphism class for each pair $(a,b)$. Distinct pairs give nonisomorphic graphs: the Petersen graph is cubic, so the degrees of the vertices adjacent to a leaf are $3+a$ and $3+b$ when $b \geq 1$, and the single value $3+a$ when $b=0$. These are the classes in $\mathcal{P}(n)$, and they attain deficit $n-10$.

For the deficit $5+\mathbf{1}_{\{n \text{ even}\}}$, use $F_2$ at odd orders and $F_0'$ at even orders. In the descriptions preceding Lemma 5.2, each of these graphs consists of internally disjoint three-edge paths between two hubs, together with one additional two-edge path in the case of $F_2$. Every cycle uses two of those paths and therefore has length at least five. An induced path that contains at most one hub meets that hub in at most two of the paths and so has at most five vertices. An induced path that contains both hubs uses at most one path between them, a segment of at most three edges, and each end extends by at most one further edge before the next vertex meets the opposite hub and creates a chord. Its order is at most six. The table of Lemma 5.2 gives deficits 5 and 6 respectively. The smaller of the two deficits is therefore attained.

*Equality.* Write $\delta(n)$ for the minimum in the statement, and let $G$ satisfy $3n-2e(G)=\delta(n)$. The identity $3n-2e(G)=\eta(Q)+n-h$ and the upper bound force the core $Q$ to be the Petersen graph, $F_2$ or $F_0'$. Every other core in the table of Lemma 5.2, and each of $K_1$ and $K_2$, has $\eta(Q)-h \geq -8$, hence deficit at least $n-8$. For $n \geq 10$ one has $n-8 > \delta(n)$, since $\delta(n) \leq n-10$.

If $Q$ is the Petersen graph, then $\eta(Q)=0$, $h=10$ and the deficit is $n-10$. This equals $\delta(n)$ if and only if $10 \leq n \leq 16$. The vertices outside $Q$ form trees attached at single vertices of $Q$. A tree of depth at least two contains a path $y,a,s$ of length two ending at its attachment vertex $s$. Concatenating that path with an induced $P_5$ of $Q$ that starts at $s$ produces an induced $P_7$: the only edge from $\{y,a\}$ into $Q$ is $as$, and $a$ is not adjacent to the neighbour of $s$ on the $P_5$ because that neighbour and $a$ would form a triangle with $s$. Thus every attached tree is a star of leaves at its attachment vertex.

The set $A$ of vertices of $Q$ that are adjacent to at least one leaf is a clique. Suppose $u,v \in A$ are distinct and nonadjacent, and let $\ell_u$ and $\ell_v$ be leaves at $u$ and at $v$. The paragraph on attainment supplies an induced $P_5$ of $Q$ with ends $u$ and $v$; write it $u=q_0,q_1,q_2,q_3,q_4=v$. The path $\ell_u,q_0,q_1,q_2,q_3,q_4,\ell_v$ has seven vertices. It is induced. The leaf $\ell_u$ is adjacent only to $q_0$, hence to none of $q_1,q_2,q_3,q_4,\ell_v$. The leaf $\ell_v$ is adjacent only to $q_4$, hence to none of $\ell_u,q_0,q_1,q_2,q_3$. The path $q_0q_1q_2q_3q_4$ is induced, so it contributes no chord. This is an induced $P_7$, which is forbidden. Thus no two vertices of $A$ are nonadjacent, and $A$ is a clique. The Petersen graph is triangle-free, so a clique has size at most two. If $|A|=2$, the two vertices are adjacent. The attachments therefore occupy a single vertex or the two ends of one edge. These are the classes in $\mathcal{P}(n)$.

If $Q=F_2$, then $\eta(Q)=5$ and $h=5+2t$ is odd, so the deficit is $5+n-h$. For odd $n$ this equals $\delta(n)=5$ if and only if $n=h$ and $n \geq 15$, in which case $G$ is $F_2$. For even $n$ the orders $n$ and $h$ have opposite parity, so $n-h \geq 1$ and the deficit is at least 6. Equality with $\delta(n)$ holds if and only if $n-h=1$ and $\delta(n)=6$, hence $n \geq 16$ is even and $G$ is $F_2$ of order $n-1$ with a single leaf. Here $t=(n-6)/2 \geq 5$.

If $Q=F_0'$, then $\eta(Q)=6$ and the deficit is $6+n-h \geq 6$, with equality if and only if $n=h$. Then $\delta(n)=6$, so $n \geq 16$ is even and $G$ is $F_0'$.

It remains to decide which vertices of $F_2$ may carry the leaf. Adding a leaf at a vertex $v$ of a $P_7$-free graph preserves $P_7$-freeness if and only if $v$ is not an end of an induced $P_6$: the leaf has degree one, so it is an end of every induced path that contains it, and the remainder is an induced path starting at $v$. Label $F_2$ as in the description before Lemma 5.2. The vertex set consists of the hubs $v_1$ and $v_3$, the midpoint $v_2$ of the unique 2-edge branch $v_1v_2v_3$, and the internal vertices of the long branches. The long branches are the $t$ matching paths through $A_1 \cup A_3$ and the path $v_1v_5v_4v_3$. The map that swaps

$v_1$ with $v_3$, swaps $v_5$ with $v_4$, fixes $v_2$, and swaps the two ends of each matching edge preserves the cycle, the matching and every spoke, so it is an automorphism. It exchanges the two hubs.
No induced $P_6$ starts at a hub. Let $P$ be an induced path starting at $v_1$. If the second vertex is $v_2$, the third is $v_3$, and the fourth is either $v_4$ or a vertex of $A_3$. In the first subcase the only continuation is $v_5$, and $v_1v_5$ is a chord, so the induced path has order at most four. In the second the only continuation is the match in $A_1$, and that match is adjacent to $v_1$, so again the order is at most four. If instead the path leaves $v_1$ along a long branch, it may continue to $v_3$ and then to the $v_3$-end of one further long branch. The resulting path has order five. Its last vertex has no neighbour off that path except its partner on the same long branch, and that partner is adjacent to $v_1$. The path does not extend, and the edge $v_1v_2$ is a chord of any continuation through $v_2$. Thus no induced $P_6$ starts at $v_1$. The automorphism gives the same conclusion at $v_3$.
No induced $P_6$ starts at $v_2$. Every induced path that starts at $v_2$ has second vertex $v_1$ or $v_3$. From $v_1$ it can enter only one long branch, and the far end of that branch is $v_3$, which is adjacent to $v_2$. The path stops before $v_3$ and has order at most four.
Every other vertex is an end of an induced $P_6$. The remaining vertices are the internal vertices of the long branches, namely $A_1 \cup A_3 \cup \{v_4, v_5\}$. Let $p$ be such a vertex and adjacent to $v_1$, so $p \in A_1$ or $p = v_5$. The automorphism carries these vertices to the vertices adjacent to $v_3$. Since $t \geq 5$, at least two long branches avoid $p$. Choose two matching branches $v_1abv_3$ and $v_1cdv_3$ among them, with $a, c \in A_1$ and with $b, d \in A_3$ their matches, and consider the path $p, v_1, a, b, v_3, d$. The edges $pv_1$, $v_1a$, $ab$, $bv_3$ and $v_3d$ are present by construction. There is no chord. A vertex of $A_1$ is adjacent only to $v_1$ and its match, a vertex of $A_3$ is adjacent only to $v_3$ and its match, $A_1$ and $A_3$ are stable sets, and the hubs are nonadjacent. If $p \in A_1$, its match is neither $b$ nor $d$, so $p$ meets the path only at $v_1$. If $p = v_5$, the only neighbours of $v_5$ are $v_1$ and $v_4$, and $v_4$ is not on the path, so again $p$ meets the path only at $v_1$. The vertex $a$ meets the path only at $v_1$ and $b$, and $b$ meets the path only at $a$ and $v_3$. Thus the path is an induced $P_6$, and a leaf at $p$ extends it to an induced $P_7$.
A leaf at a hub or at $v_2$ therefore keeps the graph $P_7$-free, and a leaf at any other vertex does not. These are the only legal positions. The two hubs are equivalent under the automorphism, while $v_2$ is the unique common neighbour of the hubs, so the two placements are not isomorphic. In $F_2$ the hub $v_1$ has degree $2 + t$, because it is adjacent to $v_2$, to $v_5$ and to the $t$ vertices of $A_1$, and $v_2$ has degree 2. After the leaf is added, the neighbour of the leaf has degree $t + 3$ or degree 3 according as it is a hub or $v_2$. For $t \geq 5$ these degrees differ.
The graphs that meet $\delta(n)$ are pairwise nonisomorphic. They are separated by the number of vertices of degree one, together with the multiset of degrees of the vertices adjacent to a leaf. A graph with a single leaf is separated by the degree of the neighbour of that leaf. For a pair $(a, b)$ those degrees are $3 + a$ and, when $b \geq 1$, $3 + b$. The graphs $F_2$ and $F_0'$ have minimum degree two. On a one-leaf extension of $F_2$ the neighbour of the leaf has degree $t + 3$ or degree 3, and these differ for $t \geq 5$. The order-15 core is $F_2$ with $t = 5$. At order 16 the underlying core has order 15, so that degree is 8 or 3. At even order $n \geq 18$ the underlying core is $F_2$ of order $n - 1$, and the degree is $n/2$ or 3. Table 1 is the resulting count. The order-15 and order-16 rows are the families drawn in Figure 4. □

| $n$ | Equality graphs | Leaves | Degrees at the leaf support | Classes |
|---|---|---|---|---|
| $10 \leq n \leq 14$ | $\mathcal{P}(n)$, pairs $(a, b)$ | $n - 10$ | none ($n = 10$); else $\{3 + a\}$ or $\{3 + a,\ 3 + b\}$ | one per pair |
| 15 | $\mathcal{P}(15)$: $(5, 0)$, $(4, 1)$, $(3, 2)$; and $F_2$, $t = 5$ | 5 or 0 | {8}, {7, 4}, {6, 5}, or none | 4 |
| 16 | $\mathcal{P}(16)$: $(6, 0)$, $(5, 1)$, $(4, 2)$, $(3, 3)$; $F_0'$; two one-leaf graphs | 6, 0 or 1 | {9}, {8, 4}, {7, 5}, {6, 6}; none; 8 or 3 | 7 |
| odd $n \geq 17$ | $F_2$ | 0 | none | 1 |
| even $n \geq 18$ | $F_0'$ and the two one-leaf graphs | 0 or 1 | none, or $n/2$, or 3 | 3 |

**TABLE 1** | Separation of the equality graphs of Theorem 5.6 by the number of leaves and the degrees of the vertices adjacent to a leaf.

For $2 \leq n \leq 9$ the same identity $3n - 2e(G) = \eta(Q) + n - h$ gives

$$f(2), f(3), \ldots, f(9) = 2, 4, 6, 10, 12, 16, 20, 24.$$

For $n \leq 4$ every connected member is a tree, so $2e = 2n - 2$. The later values are attained by $C_5$, by both $C_6$ and $C_5$ with one leaf, by $F_2$ with $t = 1$, by $F_3$ and by $F_4$. The graphs $C_5$, $C_6$, $F_2$ with $t = 1$, $F_3$ and $F_4$ lie on the list of Lemma 5.2. The graph $C_5$ with one leaf is that core with a single attached leaf, as in Lemma 5.4. Every other core of order at most $n$, after the attachment of $n - h$ vertices, has weakly smaller $2e$.

**Theorem 5.7.** *Let $G$ be a connected strong $C_4$-free $P_7$-free graph with at most one triangle. Then $S(G) \leq 3n$.*

*Proof.* Suppose $G$ is a counterexample of minimum order in the stated class, so $S(G) \geq 3n + 1$. Theorem 5.5 forces $G$ to contain a triangle, hence exactly one. Let $v$ be a cut vertex. For each component $C$ of $G - v$ let $\rho_C$ be the largest order of an

induced path in $G[C\cup\{v\}]$ with end $v$. Two such paths from different components form an induced path on $\rho_C+\rho_{C'}-1\le 6$ vertices, so some component $C_1$ has $\rho_{C_1}\le 3$. Put $G_1=G[C_1\cup\{v\}]$ and $G_2=G-C_1$, with $n_i=|G_i|$, so $n=n_1+n_2-1$ and $n_1\ge 2$. Both graphs are connected, smaller than $G$, strong $C_4$-free and $P_7$-free, and each has at most one triangle. Lemma 5.3 gives $S(G_1)\le 2(n_1-1)$, and minimality gives $S(G_2)\le 3n_2$. By Lemma 2.3,

$$S(G)\le 2(n_1-1)+3n_2=3n-(n_1-1)<3n.$$

Thus $G$ is 2-connected.

Let $v$ be a vertex of minimum degree among the three vertices of the triangle, write $d=d(v)$, and let $u$ and $w$ be the other two. The graph $H=G-v$ is connected and triangle-free. Apply Lemma 5.4 to $H$, and write $Q$ for its core, $h=|V(Q)|$ and $\eta=\eta(Q)$. Set

$$a=\bigl|N_G(v)\cap V(Q)\bigr|,\qquad c=\bigl|V(H)\setminus\bigl(V(Q)\cup N_G(v)\bigr)\bigr|,$$

and $t=|V(H)|-h$. The vertices of $H$ outside $Q$ are the $d-a$ neighbours of $v$ that miss $Q$, together with the $c$ vertices counted above, so $t=d-a+c$. Lemma 5.4 gives $e(H)=e(Q)+t$. Since $G$ has one triangle, $S(G)=2e(G)-3$ and $e(G)=e(H)+d$, and therefore

$$S(G)-3n=2d-6-\eta-t=d+a-6-\eta-c. \tag{5.1}$$

The core satisfies $a\le\eta+3$ and, for every edge $xy$ of $Q$,

$$a+\min\{d_Q(x),d_Q(y)\}\le\eta+5. \tag{5.2}$$

These bounds are read from the list in Lemma 5.2, using the descriptions that precede that lemma. If $Q$ is the Petersen graph, then $\eta=0$. Every two nonadjacent vertices of the Petersen graph have a common neighbour, so two such vertices cannot both be adjacent to $v$. Thus $N(v)\cap V(Q)$ is a clique, necessarily of size at most two, and every degree in $Q$ is three. Both bounds follow. For each remaining fixed graph of order at least three one has $h\le 11$ and minimum degree at least two. The open neighbourhoods in $Q$ of any two vertices of $N(v)\cap V(Q)$ are disjoint: a common neighbour would form a 4-cycle with $v$. Hence $a\le\lfloor h/2\rfloor\le 5$. Every degree in these graphs is at most four. The only rows with $\eta<4$ are $F_4$ and $F_{12}$, both with $\eta=3$. The path-and-spoke description of $F_4$ has maximum degree three. In $F_{12}$ the only vertices of degree four are $v_1$ and $v_5$ on the 7-cycle. Each is adjacent to two cycle vertices and to two of $\{x_1,x_2,x_5\}$, and $v_1$ is not adjacent to $v_5$, so every edge of $F_{12}$ has an end of degree at most three. This gives (5.2) and the bound on $a$. In each parameterised family every non-hub vertex is adjacent to a hub, and each hub has at most one neighbour in common with $v$, so $a\le 4$ even after both hubs are counted. Every edge has an end of degree two, because the edges are the internal edges of the paths between the hubs described above, and $\eta$ is 5 or 6. The graphs $K_1$ and $K_2$ satisfy both bounds directly: $K_1$ has $\eta=3$ and no edge, and $K_2$ has $\eta=4$.

If $u$ or $w$ lies outside $Q$, say $u$, the tree attachments give $u$ at most one neighbour in $Q$. Apart from $w$, no neighbour of $u$ in $H$ is adjacent to $v$, because the triangle is unique. Thus $d\le d(u)\le c+3$. Substituting in (5.1) gives $S(G)-3n\le a-\eta-3$, and $a\le\eta+3$ yields $S(G)-3n\le 0$. If instead both $u$ and $w$ lie in $Q$, then $uw$ is an edge of $Q$, and every neighbour of either vertex that lies in $H-V(Q)$ lies in the set counted by $c$. Since $v$ has minimum degree among the three triangle vertices,

$$d\le 1+c+\min\{d_Q(u),d_Q(w)\}.$$

Substituting in (5.1) and applying (5.2) to the edge $uw$ again yields $S(G)\le 3n$. □

**Lemma 5.8.** *Let $G$ be strong $C_4$-free, and let $T_1$ and $T_2$ be triangles of $G$.*

1. *Suppose $T_1$ and $T_2$ share a vertex $u$, and write $T_1=\{u,a,b\}$ and $T_2=\{u,c,d\}$. If $ac$ is an edge, then $a\,c\,d\,u$ is a 4-cycle. The same holds for each of the other three edges with one end in $\{a,b\}$ and one end in $\{c,d\}$.*
2. *Suppose $T_1$ and $T_2$ are disjoint, and let $xy$ and $x'y'$ be distinct edges with $x,x'\in T_1$ and $y,y'\in T_2$. If the two edges share an endpoint in $T_1$, say $x=x'$ and $y\ne y'$, and if $t$ is the third vertex of $T_2$, then $x\,y\,t\,y'$ is a 4-cycle. If they share an endpoint in $T_2$, the same construction with the roles of $T_1$ and $T_2$ exchanged is a 4-cycle. If they share no endpoint, then $x\ne x'$ and $y\ne y'$, both pairs are edges of their triangles, and $x\,y\,y'\,x'$ is a 4-cycle. In particular there is at most one edge with one end in $T_1$ and one end in $T_2$.*
3. *Suppose $T_1$ and $T_2$ are disjoint and lie in the same component. Let $d$ be the distance between their vertex sets, and let $P:u=p_0,p_1,\dots,p_d=v$ be a path of length $d$ with $u\in T_1$ and $v\in T_2$. The internal vertices of $P$ lie outside $T_1\cup T_2$, and the subgraph induced by $V(T_1)\cup V(T_2)\cup V(P)$ consists of the edges of $T_1$, the edges of $T_2$ and the edges of $P$.*
4. *In the notation of (3), if $x\in T_1\setminus\{u\}$ and $z\in T_2\setminus\{v\}$, then $x,u,p_1,\dots,p_{d-1},v,z$ is an induced path on $d+3$ vertices.*

*Proof.* (1) The vertices $a, c, d$ and $u$ are distinct: $T_1$ and $T_2$ share only $u$, so $a \notin T_2$. The edges $ac$, $cd$, $du$ and $ua$ are present, the last three because $cd$ and $du$ lie in $T_2$ and $ua$ lies in $T_1$. This is the 4-cycle $a\,c\,d\,u$. Each of the other three cross edges is the same configuration after renaming.
(2) First suppose $x = x'$ and $y \neq y'$. Then $y$ and $y'$ are distinct vertices of the triangle $T_2$, so both are adjacent to the third vertex $t$ of $T_2$. The vertices $x, y, t, y'$ are distinct because $x \notin T_2$. The edges $xy$, $yt$, $ty'$ and $y'x$ are present, and $x\,y\,t\,y'$ is a 4-cycle. The case of a shared endpoint in $T_2$ is the same argument with the triangles exchanged.
Now suppose the four endpoints are distinct. Any two vertices of a triangle are adjacent, so $xx'$ and $yy'$ are edges. The vertices $x, y, y', x'$ are distinct because the triangles are disjoint. The edges $xy$, $yy'$, $y'x'$ and $x'x$ are present, and $x\,y\,y'\,x'$ is a 4-cycle. A second edge between $T_1$ and $T_2$ falls under one of these two cases, so a strong $C_4$-free graph has at most one such edge.
(3) A shortest path between the two sets meets each set only at an endpoint, so every internal vertex lies outside $T_1 \cup T_2$. A chord of $P$ would give a shorter $u$–$v$ path.
Let $x \in T_1 \setminus \{u\}$, and let $y$ be the third vertex of $T_1$, so $\{u, x, y\}$ is a triangle. Suppose $x$ is adjacent to $p_i$ for some $i \in \{1, \dots, d\}$. If $i = 1$, the vertices $y, u, p_1, x$ are distinct and the edges $yu$, $up_1$, $p_1x$ and $xy$ are present, so $y\,u\,p_1\,x$ is a 4-cycle. If $i \geq 2$, the walk $x, p_i, p_{i+1}, \dots, p_d$ has length $d - i + 1 \leq d - 1$ and joins $T_1$ to $T_2$, contradicting the definition of $d$. Thus $x$ has no neighbour on $P$ except $u$. The same argument, using the third vertex of $T_2$, shows that no vertex of $T_2 \setminus \{v\}$ has a neighbour on $P$ except $v$. In particular, adjacency to $p_{d-1}$ produces a 4-cycle, and adjacency to an earlier vertex of $P$ shortens the distance.
It remains to rule out an edge with one end in $T_1$ and one end in $T_2$, other than the edge $uv$ in the case $d = 1$. If $d \geq 2$, any such edge would be a path of length 1 between the two sets. If $d = 1$, the path $P$ is the edge $uv$, and a second edge is a 4-cycle by (2).
No other pair of vertices in $V(T_1) \cup V(T_2) \cup V(P)$ remains. The induced subgraph is therefore the two triangles joined by $P$.
(4) The named walk has $d + 3$ vertices and uses edges of $T_1$, of $P$ and of $T_2$. A chord incident with $x$ would be an edge from $x$ to a vertex of the walk other than $u$. Part (3) forbids every such edge, including an edge from $x$ to $z$. A chord incident with $z$ is forbidden in the same way. A chord that avoids $x$ and $z$ is a chord of $P$ or an edge forbidden by (3). The walk is an induced path. □

**Lemma 5.9.** *Every connected strong $C_4$-free $P_7$-free graph with at least two triangles has a vertex $v$ with $L(v) \leq 3$.*

*Proof.* Let $G$ be connected, strong $C_4$-free and $P_7$-free, and suppose $G$ has at least two triangles. Two triangles do not share an edge. Suppose they share the edge $uv$, and let $x$ and $y$ be the third vertices. The vertices $x, u, y, v$ are distinct, and the edges $xu$, $uy$, $yv$ and $vx$ are present, because each of $x$ and $y$ is adjacent to both ends of $uv$. The cycle $x\,u\,y\,v$ is a 4-cycle. Thus two triangles share at most one vertex.
If two triangles share a vertex, Lemma 5.8(1) forbids an edge between their unshared vertices. Their union is an induced subgraph on five vertices. If the triangles are disjoint, let $d$ be the distance between their vertex sets. Lemma 5.8(3) identifies the induced subgraph formed by the two triangles and a shortest connecting path, and Lemma 5.8(4) supplies an induced path on $d + 3$ vertices. Since $p(G) \leq 6$, one has $d \leq 3$.
Four induced subgraphs therefore exhaust the ways two triangles can meet. They may share a vertex, or they may be joined by a path of length 1, 2 or 3. These graphs have orders 5, 6, 7 and 8. Call them the seeds, and label the vertices of a seed by 0, 1, ... in the order used by Appendix A. From a seed, grow a labelled graph $H$ as follows. Let $X$ be the set of vertices $v$ with $L_H(v) < 4$. A node is unrestricted when $X$ is empty, that is, when every vertex of $H$ has deletion loss at least 4. While $X$ is nonempty, choose the vertex $v \in X$ with the fewest legal children, and if several vertices have that number, choose the smallest label. A legal child is obtained by adding one vertex whose neighbourhood $U \subseteq V(H)$ contains $v$, provided the enlarged graph has no 4-cycle subgraph and no induced $P_7$. A 4-cycle through the new vertex exists precisely when two vertices of $U$ have a common neighbour in $H$. The children are explored in increasing order of the integer bitmask of $U$. There is no order cutoff and no isomorphism pruning. The four searches visit 36, 31, 46 and 5 nodes, of which 7, 6, 21 and 3 are leaves, and the largest orders they reach are 17, 15, 11 and 10. None of the four searches has an unrestricted node. Every node has $X$ nonempty, and every branch ends at a leaf. Appendix A prints the 118 decisions, one entry per node, and records the induced-$P_7$ test. The same arrays are produced by `referee_checks/lemma57b.c`, the re-implementation supplied with a JGT referee report (2026), run with deficient-vertex decisions only.
Suppose $G$ is a connected strong $C_4$-free $P_7$-free graph with at least two triangles and with $L \geq 4$ at every vertex. Then $G$ contains one of the four seeds as an induced subgraph. Each search tree is finite and has no unrestricted node, so the branch that follows $G$ ends at a leaf $H$. At that leaf the chosen vertex $v$ has deletion loss $L_H(v) < 4$ and no legal new neighbour. The loss on an induced subgraph is $L = 2d - 3t$. If $v$ had no neighbour in $G$ outside $H$, then, because $H$ is induced, the degree and the triangle count of $v$ would already be final, and $L_G(v) = L_H(v) < 4$, contradicting the choice of $G$. Thus $G$ supplies a neighbour $z$ of $v$ outside $H$. The neighbourhood $U = N(z) \cap V(H)$ contains $v$, and the graph formed by $H$ and $z$ is an induced subgraph of $G$, so it has no 4-cycle subgraph and no induced $P_7$. It is a legal child of the

leaf, which is impossible. Hence no such graph $G$ exists. The same conclusion holds for any selection rule that, whenever $X$ is nonempty, chooses some vertex of $X$ and explores every legal neighbourhood containing it. Fail-first is the rule that produced the arrays in Appendix A. Every connected strong $C_4$-free $P_7$-free graph with at least two triangles has a vertex of deletion loss at most 3. □

**Theorem 5.10.** *Every connected strong $C_4$-free $P_7$-free graph $G$ satisfies $S(G) \le 3n$, with equality if and only if $G$ is the Petersen graph.*

*Proof.* The triangle-free case, including the equality, is Theorem 5.5. Suppose $G$ contains a triangle and $S(G) \ge 3n$, and choose such a connected graph of minimum order. Every smaller connected graph in the class satisfies $S \le 3|V|$, and the inequality is strict whenever the smaller graph contains a triangle.

Suppose $s$ is a cut vertex. Induced paths that start at $s$ in distinct components of $G - s$ concatenate. Since $p(G) \le 6$, some component determines a side $G_1$, induced by that component together with $s$, in which every induced path starting at $s$ has order at most three. Let $G_2$ be the subgraph induced by the remaining vertices together with $s$, and write $n_i = |V(G_i)|$. Then $n = n_1 + n_2 - 1$ and $n_1 \ge 2$. Lemma 5.3 gives $S(G_1) \le 2(n_1 - 1)$. The graph $G_2$ is a smaller connected graph in the class. If it contains a triangle, minimality gives $S(G_2) < 3n_2$; if it does not, Theorem 5.5 gives $S(G_2) \le 3n_2$. Lemma 2.3 therefore yields

$$S(G) \le 2(n_1 - 1) + 3n_2 = 3n - (n_1 - 1) < 3n.$$

Thus $G$ has no cut vertex, and $G - v$ is connected for every vertex $v$.

Suppose $L(v) \le 3$ for some vertex $v$. Then $S(G - v) = S(G) - L(v) \ge 3n - 3 = 3(n - 1)$. The graph $G - v$ is connected and lies in the class. It contains no triangle: otherwise minimality would give the strict inequality $S(G - v) < 3(n - 1)$. Theorem 5.5 therefore forces $G - v$ to be the Petersen graph, so $n = 11$ and $S(G - v) = 30$. Any two nonadjacent vertices of the Petersen graph have a common neighbour, so the neighbours of $v$ form a clique. The Petersen graph is triangle-free, and a clique in it has size at most two. The graph $G$ contains a triangle, so $v$ is adjacent to both ends of an edge and to no further vertex of $G - v$. Then $d(v) = 2$, $t(v) = 1$ and $L(v) = 2 \cdot 2 - 3 \cdot 1 = 1$, hence $S(G) = 31 < 33$. This contradiction shows that $L(v) \ge 4$ for every vertex $v$.

Suppose $u$ and $w$ are adjacent vertices of degree two. The bound on the deletion loss forces $t(u) = t(w) = 0$, so neither vertex lies in a triangle and the two neighbours of $u$ are nonadjacent. Let $x$ be the neighbour of $u$ other than $w$, and let $y$ be the neighbour of $w$ other than $u$. Then $x \ne y$, the edge $xy$ is absent, and $x$ is not adjacent to $w$. Deleting $\{u, w\}$ removes the three edges $xu$, $uw$ and $wy$ and preserves every triangle. The graph that remains may be disconnected. Its number of edges is $e(G) - 3$ and its number of triangles is $T(G)$, so its value of $S$ is $S(G) - 6 \ge 3n - 6 = 3(n - 2)$. For a disconnected graph, $S$ is the sum of the values on the components. Each component is a smaller connected graph in the class. The triangle of $G$ survives in one component, and minimality makes the inequality strict for that component, while every other component contributes at most three times its order. The sum is therefore strictly less than $3(n - 2)$, a contradiction.

Thus $L(v) \ge 4$ at every vertex and no two vertices of degree two are adjacent. If $T(G) \ge 2$, Lemma 5.9 supplies a vertex of deletion loss at most 3, a contradiction. Hence $T(G) = 1$. Theorem 5.7 gives $S(G) \le 3n$, so $S(G) = 3n$.

Let $v$ be a vertex of minimum degree on the unique triangle, write $d = d(v)$, and let $u$ and $w$ be the other two vertices. The graph $H = G - v$ is connected and triangle-free. Let $Q$ be its core under Lemma 5.4, of order $h$ and deficit $\eta = \eta(Q)$, and define $a$ and $c$ as in the proof of Theorem 5.7. That proof gives the identity (5.1) and the bounds $a \le \eta + 3$ and (5.2). The same case division yields the sharper bound $a \le \eta + 2$. If $Q$ is the Petersen graph then $a \le 2 = \eta + 2$. For each remaining fixed graph, $a \le \lfloor h/2 \rfloor \le 5$, while the only rows of Lemma 5.2 with $\eta < 4$ are $F_4$ and $F_{12}$, where the descriptions give $a \le 4$ and $a \le 5$ respectively. Every other fixed row has $\eta \ge 4$, so $\eta + 2 \ge 6$. In each parameterised family $a \le 4$ and $\eta \in \{5, 6\}$. The graph $K_1$ has $a \le 1$ and $\eta = 3$, and $K_2$ has $a \le 2$ and $\eta = 4$.

If $u$ or $w$ lies outside $Q$, the estimate in the proof of Theorem 5.7 becomes $S(G) - 3n \le a - \eta - 3 \le -1$. Thus both $u$ and $w$ lie in $Q$. The same proof then gives $d \le 1 + c + \min\{d_Q(u), d_Q(w)\}$, and

$$S(G) - 3n \le a + \min\{d_Q(u), d_Q(w)\} - \eta - 5.$$

Equality in (5.1) forces both

$$d = 1 + c + \min\{d_Q(u), d_Q(w)\} \qquad \text{and} \qquad a + \min\{d_Q(u), d_Q(w)\} = \eta + 5. \tag{5.3}$$

No core other than the Petersen graph meets (5.3). In the parameterised families every edge has an end of degree two and $a \le 4$, so the left-hand sum is at most 6, while $\eta + 5 \ge 10$. On $K_2$ the sum is at most 3. On $C_5$, $C_6$ and $C_7$ every degree is 2 and $\lfloor h/2 \rfloor + 2 \le 5 < \eta + 5$. The graph $F_4$ has maximum degree three and $a \le 4$, so the sum is at most $7 < 8$. On $F_3$ the three added vertices induce a path and meet three distinct cycle vertices, so the maximum degree is three and, with $a \le 4$, the sum is at most $7 < 9$. On $F_7$ and on $F_8$ each added vertex meets the 7-cycle in two vertices, the traces are disjoint,

and the maximum degree is three; the resulting sums are at most 7, against $\eta + 5 \in \{11, 10\}$. On $F_{10}$ the only vertex of degree four is $v_1$, each of whose four neighbours has degree two, so every edge has an end of degree at most three and, with $a \leq 4$, the sum is at most $7 < 10$. On $F_9$, obtained from $F_8$ by a vertex adjacent only to $x_1$ and $x_2$, the maximum degree is three and the sum is at most $8 < 9$. On $F_{11}$ the vertices of degree four are $v_1$ and $v_5$, which are nonadjacent on the 7-cycle, and every neighbour of either has degree at most three, so every edge has an end of degree at most three and the sum is at most $8 < 9$. On $F_{12}$ every edge has an end of degree at most three and $a \leq 5$, so equality would require $a = 5$ and $\min\{d_Q(u), d_Q(w)\} = 3$. The five vertices of $N(v) \cap V(Q)$ have pairwise disjoint neighbourhoods in a graph of order 11 and minimum degree two. The edge $uw$ contributes degrees adding to at least $3 + 3$, and the other three vertices contribute at least 2 each, so the sum of the five degrees is at least $12 > 11$.

Thus $Q$ is the Petersen graph, $\eta = 0$, $a = 2$ and $\min\{d_Q(u), d_Q(w)\} = 3$. Every degree in $Q$ is three, so $d_Q(u) = d_Q(w) = 3$. Write $c_u$ for the number of neighbours of $u$ in $H - V(Q)$, and $c_w$ likewise. A common neighbour would form a triangle with the edge $uw$, so these neighbourhoods are disjoint and $c_u + c_w \leq c$. As in the proof of Theorem 5.7, $d(u) \leq 1 + d_Q(u) + c_u$ and $d(w) \leq 1 + d_Q(w) + c_w$. Since $d$ is at most both $d(u)$ and $d(w)$, the equality $d = 4 + c$ forces $c \leq c_u$ and $c \leq c_w$. Combined with $c_u + c_w \leq c$ this gives $c = 0$, and then $d = 4$. The integer $t = d - a$ equals 2, so $H$ consists of $Q$ and two further vertices, both adjacent to $v$.

Those two vertices are leaves of $H$, attached at distinct vertices $q_1$ and $q_2$ of $Q$. A path of length two in an attached tree would place both of its vertices in $N(v)$, and the edge between them would form a second triangle with $v$. The supports are distinct, for a common support would give that vertex and $v$ two common neighbours. Neither support is $u$ or $w$, because $c_u = c_w = 0$. Neither support is adjacent to $u$ or to $w$: the edge from $u$ to $q_1$, together with the leaf at $q_1$ and the edge $uv$, would be a 4-cycle.

If $q_1$ and $q_2$ are nonadjacent, the proof of Theorem 5.6 supplies an induced $P_5$ of $Q$ with those ends. The two leaves are nonadjacent, since both are adjacent to $v$ and the triangle is unique, and each leaf meets $Q$ only at its support. Adding the leaves to the $P_5$ produces an induced $P_7$.

If $q_1$ and $q_2$ are adjacent, realise $Q$ on the 2-element subsets of $\{1, 2, 3, 4, 5\}$, adjacent when disjoint, as in the proof of Theorem 5.6. The symmetric group on the ground set acts by graph automorphisms and is edge-transitive, so it carries the edge $uw$ to $\{\{1, 2\}, \{3, 4\}\}$. The neighbours of $\{1, 2\}$ are the three pairs in $\{3, 4, 5\}$, and the neighbours of $\{3, 4\}$ are $\{1, 2\}$, $\{1, 5\}$ and $\{2, 5\}$. The four remaining vertices are $\{1, 3\}$, $\{1, 4\}$, $\{2, 3\}$ and $\{2, 4\}$, and the only edges among them are $\{1, 3\}\{2, 4\}$ and $\{1, 4\}\{2, 3\}$. The transposition (1 2) fixes $uw$ and exchanges those two edges, so in one such labelling $\{q_1, q_2\} = \{\{1, 3\}, \{2, 4\}\}$. Let $x$ be the leaf at $\{1, 3\}$. The path

$$v,\ x,\ \{1, 3\},\ \{2, 4\},\ \{1, 5\},\ \{2, 3\},\ \{1, 4\}$$

is an induced $P_7$. Consecutive vertices are adjacent by the construction of $G$ or by disjointness. The vertex $v$ meets $Q$ only in $\{1, 2\}$ and $\{3, 4\}$, the leaf $x$ meets $Q$ only in $\{1, 3\}$, and each of $\{1, 3\}$, $\{2, 4\}$, $\{1, 5\}$ and $\{2, 3\}$ meets every later nonconsecutive vertex of the path in a pair that shares an element of $\{1, 2, 3, 4, 5\}$.

In every case $G$ contains an induced $P_7$. This contradiction shows that no connected graph in the class with a triangle satisfies $S \geq 3n$. Combined with Theorem 5.5, one has $S \leq 3n$, with equality only for the Petersen graph. □

**Theorem 5.11.** *Let $G$ be a connected graph on at least two vertices with $\ell(G) \leq 4$. Then*

$$p(G) \geq r(G) + \ell(G)^{c_{C_4}(G)}.$$

*If $c_{C_4}(G) = 0$, the power equals 1. The conjecture remains open for graphs with $\ell > 4$.*

*Proof.* If $G$ contains a 4-cycle, then $c_{C_4}(G) = 0$ and the inequality is $p(G) \geq r(G) + 1$. A shortest path between two vertices at distance $\operatorname{diam}(G) \geq r(G)$ is induced.

Suppose now that $G$ is strong $C_4$-free, and write $r = r(G)$. The inequality is $p \geq r + \ell$.

- $r = 1$. Lemma 3.1 gives $\ell < 2$ and $p \geq r + \lceil \ell \rceil$, hence $p \geq r + \ell$.
- $r = 2$. If $\ell \leq 2$, Lemma 3.2 gives $p \geq 4$. If $2 < \ell \leq 3$ and $p \leq 4$, Lemma 3.3 gives $\ell \leq 2$, a contradiction, so $p \geq 5$. If $3 < \ell \leq 4$ and $p \leq 5$, Theorem 3.6 gives $\ell \leq 3$, a contradiction, and hence $p \geq 6$.
- $r = 3$. If $\ell \leq 2$, Theorem 2.4 gives $p \geq 5$. If $2 < \ell \leq 3$ and $p \leq 5$, then Lemma 3.5 and $r = 3$ give $G = C_6$, with $\ell = 2$, so $p \geq 6$. If $3 < \ell \leq 4$ and $p \leq 5$, Theorem 3.6 gives $\ell \leq 3$, a contradiction. If $3 < \ell \leq 4$ and $p = 6$, Theorem 5.10 gives $\ell \leq 3$, a contradiction. Hence $p \geq 7 \geq r + \ell$.
- $r = 4$. Theorem 2.4 gives $p \geq 7$. If $p = 7$, Corollary 4.4 gives $\ell \leq 2$, so $p \geq r + \ell$. If $p \geq 8$ and $\ell \leq 4$, then $p \geq 8 = r + 4 \geq r + \ell$.
- $r \geq 5$. Theorem 2.4 gives $p \geq 2r - 1 \geq r + 4 \geq r + \ell$.

□

Theorem 5.11 uses Lemmas 3.1, 3.2, 3.3 and 3.5. Through Lemma 3.5 it also uses Lemmas 3.4 and 2.8, Theorem 2.5 and Theorem 2.6. It uses Theorem 2.4; Theorem 3.6 when $r = 2$, and when $r = 3$ with $p \le 5$; Corollary 4.4 through Theorem 4.2, together with the case $r = 4$ and $p \ge 8$ above; and Theorem 5.10, through Theorems 5.5 and 5.7, Lemma 5.9 and the Chen–Xu classification. Lemma 5.9 enters only through Theorem 5.10, and only for graphs with at least two triangles. In this proof that happens only when $r = 3$, $3 < \ell \le 4$ and $p = 6$.

Since $\lambda(v) \le d(v)$ for every vertex, $\ell \le 2e/n$. Theorem 5.11 covers every connected graph of average degree at most 4. Any two vertices of a strong $C_4$-free graph have at most one common neighbour, so

$$\sum_v \binom{d(v)}{2} \le \binom{n}{2}.$$

Convexity yields $(2e/n)(2e/n - 1) \le n - 1$, hence

$$\frac{2e}{n} \le \frac{1 + \sqrt{4n - 3}}{2}.$$

This is Reiman's bound [14]; the Kővári–Sós–Turán theorem gives the same order of magnitude for subgraphs $K_{2,2}$ [11]. Since $S = 2e - 3T$, one has $\ell \le 2e/n$. For $n \le 13$ one has $\sqrt{4n - 3} \le 7$, so $\ell \le 4$. Every connected strong $C_4$-free graph on at most 13 vertices therefore falls under Theorem 5.11. Every connected graph on at most 13 vertices that contains a 4-cycle satisfies Conjecture 1.1 because the exponent is 0. The theorem covers every connected graph of order at most 13. The conjecture for $\ell > 4$ is untouched. The Erdős–Saks–Sós bound $p \ge 2r - 1$ gives $p \ge r + \ell$ whenever $\ell \le r - 1$. Once $\ell > 4$ and $p \ge 7$, Theorem 5.10 no longer applies, and the missing statement is an analogue such as $S \le 4n$ for connected strong $C_4$-free $P_9$-free graphs. Corollary 6.1 splits that open range into $\mathrm{GEN}_K$ and $\mathrm{RAD}(r)$.

Equality in the real-valued inequality holds for $K_2$ and $K_3$ ($r = 1$, $\ell = 1$, $p = 2$), for $C_5$ ($r = 2$, $\ell = 2$, $p = 4$), for the Petersen graph ($r = 2$, $\ell = 3$, $p = 5$) and for $C_6$ ($r = 3$, $\ell = 2$, $p = 5$). The 2026 referee report records that these five are the only equality cases among connected strong $C_4$-free graphs on at most 13 vertices; that enumeration is not given a proof here. The article does not characterise the equality cases in general.

*Remark 5.12.* Every connected graph $G$ with $\ell(G) < 5$ satisfies

$$p(G) \ge r(G) + \left\lfloor \ell(G)^{c_{C_4}(G)} \right\rfloor.$$

The exponent follows the convention of Conjecture 1.1: if $c_{C_4}(G) = 0$, then the power equals 1. If $G$ contains a 4-cycle, both the real form and this floor form reduce to $p \ge r + 1$, which follows from a shortest path between two vertices at distance $\mathrm{diam}(G) \ge r(G)$. If $G$ is strong $C_4$-free and $\ell \le 4$, Theorem 5.11 gives $p \ge r + \ell$, and $\ell \ge \lfloor \ell \rfloor$. Suppose $G$ is strong $C_4$-free and $4 \le \ell < 5$, and suppose the floor form fails. Then $\lfloor \ell \rfloor = 4$, so $p \le r + 3$. Theorem 2.4 gives $p \ge 2r - 1$, so $r \le 4$. If $r \le 3$, then $p \le 6$, and Theorem 5.10 gives $\ell \le 3$, contradicting $\ell \ge 4$. If $r = 4$, then $p = 7$, and Corollary 4.4 gives $\ell \le 2$, again a contradiction. The real-valued conjecture for $\ell > 4$ remains open.

## 6 | Concluding remarks and open problems

For integers $K \ge 3$ and $r \ge 4$, write $\mathrm{GEN}_K$ for the statement that every connected strong $C_4$-free graph $G$ with $p(G) \le 2K$ satisfies $S(G) \le Kn$, and write $\mathrm{RAD}(r)$ for the statement that every connected strong $C_4$-free graph with $r(G) = r$ and $p(G) = 2r - 1$ satisfies $\ell(G) \le r - 1$. Corollary 4.4 is the case $r = 4$ of $\mathrm{RAD}(r)$, in the stronger form $\ell \le 2$.

**Corollary 6.1.** *Let $G$ be a connected strong $C_4$-free graph that violates Conjecture 1.1. Then $\ell(G) > 4$. Put $K = \lceil \ell(G) \rceil - 1 \ge 4$. Then either*

*(a) $r(G) \le K$ and $p(G) \le r(G) + K \le 2K$, so $G$ violates $\mathrm{GEN}_K$; or*

*(b) $r(G) = K + 1 \ge 5$ and $p(G) = 2r(G) - 1$, and $\ell(G) > r(G) - 1$, so $G$ violates $\mathrm{RAD}(K + 1)$.*

*Proof.* Write $r = r(G)$, $p = p(G)$ and $c = \lceil \ell(G) \rceil = K + 1$. Then $\ell > 4$ by Theorem 5.11, so $K \ge 4$. A violation means $p \le r + c - 1 = r + K$. Theorem 2.4 gives $2r - 1 \le p \le r + K$, so $r \le K + 1$. If $r \le K$, then $p \le r + K \le 2K$, and $S > Kn$ because $\ell > K$. If $r = K + 1$, then $2r - 1 \le p \le 2K + 1 = 2r - 1$, and $\ell > K = r - 1$. The radius in (b) is at least 5. □

**Proposition 6.2.** *For $k \ge 3$ let $\theta_k$ consist of two vertices $x, y$ joined by $k$ internally disjoint paths $xa_ib_iy$. This is the graph $F_0'$ of Section 5 with parameter $t' = k - 2$. Then $\theta_k$ is strong $C_4$-free, $n = 2k + 2$, $r = 3$, $p = 6$ and $S = 6k = 3n - 6$. Thus $p - r - \ell = 6/n \to 0$.*

*Proof.* Every cycle consists of two complete branches and so has length 6; in particular there is no 4-cycle. There are no triangles, so $S = 2e = 6k$. Every vertex has eccentricity 3: $d(x, y) = 3$ and $d(a_i, b_j) = 3$ for $i \neq j$. The path $a_j x a_i b_i y b_l$ (with $j, i, l$ distinct) is induced. Conversely, let $P$ be an induced path. If $P$ avoids $x$ and $y$, it lies inside one edge $a_i b_i$. If $P$ contains $x$ but not $y$, then $x$ has at most two neighbours on $P$, each followed on $P$ by at most its $b$-vertex, so $|P| \leq 5$. If $P$ contains $x$ and $y$, then every $a_j$ on $P$ is adjacent on $P$ to $x$ and every $b_j$ to $y$, at most one branch lies entirely on $P$, and so $P$ consists of $x$, $y$, at most one full branch and at most one further neighbour of each of $x$ and $y$. Hence $|P| \leq 6$. □

**Corollary 6.3.** *Let $G$ be a connected strong $C_4$-free graph with $p(G) = 6$. Then $S(G) \leq 3n - 1$ and $\ell(G) < 3$. On each of the classes $(r, p) = (2, 6)$ and $(r, p) = (3, 6)$ the supremum of $\ell$ is 3, and the supremum is not attained.*

*Proof.* The graph is $P_7$-free. Theorem 5.10 gives $S(G) \leq 3n$, with equality only for the Petersen graph. The Petersen graph has induced-path order 5, so $S(G) \leq 3n - 1$ and $\ell(G) < 3$.
Both constructions below are strictly less than 3 and tend to 3 as the order tends to infinity. The supremum on each class is therefore 3, and it is not attained.
For the class $(2, 6)$, take the graph $F_2$ defined before Lemma 5.2, with parameter $t \geq 2$. In the description preceding Lemma 5.2 the hubs are $v_1$ and $v_3$, the unique branch of length 2 is $v_1 v_2 v_3$, and the remaining $v_1$–$v_3$ paths are the $t$ matching paths through $A_1 \cup A_3$ together with $v_1 v_5 v_4 v_3$. These $t + 1 \geq 3$ paths are branches of length 3. The order is $n = 5 + 2t$. The graph is triangle-free, and the table of Lemma 5.2 gives $2e = 3n - 5$, so $\ell = 3 - 5/n$. Every vertex other than the two hubs is adjacent to a hub, and $v_2$ is adjacent to both hubs, so every vertex is within distance 2 of $v_2$. A vertex of $A_1$ is not adjacent to $v_2$. Each hub has degree $t + 2 = (n - 1)/2 < n - 1$, and every other vertex has degree 2, so no vertex dominates the graph. The radius is 2. The proof of Theorem 5.6 shows that every induced path has order at most 6. For the matching lower bound, choose three branches of length 3, written $v_1 a b v_3$, $v_1 c d v_3$ and $v_1 e f v_3$. The path $a, v_1, c, d, v_3, f$ is induced: the hubs are nonadjacent, and an internal vertex of any one of these branches is adjacent only to its two neighbours on that branch. Thus $(r, p, \ell) = (2, 6, 3 - 5/n)$.
For the class $(3, 6)$, Proposition 6.2 gives the graph $\theta_k$ for every $k \geq 3$, with $\ell = 3 - 6/n$. □

The two families of Corollary 6.3 are why a uniform gap below 3 is not available.
Theorem 5.11 stops at $\ell = 4$ because that is the largest constant for which the available gap $p - r$ meets $\ell$. Theorem 5.10 gives $\ell \leq 3$ whenever $p \leq 6$. Corollary 4.4 gives $\ell \leq 2$ when $r = 4$ and $p = 7$. For $r \geq 5$, Theorem 2.4 gives $p \geq 2r - 1$, hence a gap of at least 4, which meets $\ell \leq 4$ and no larger constant. A graph with $\ell > 4$ has $p \geq 7$. The bound for induced-path order at most 6 does not apply to it, and if its radius is at least 5 the radius-four classification does not apply either. The obstruction to a larger constant is the lack of an analogue of Theorem 5.10, such as $S \leq 4n$ for every connected strong $C_4$-free $P_9$-free graph. No such bound is proved here.
Corollary 6.1 splits any counterexample to REAL-133 into two open statements: either $S > Kn$ for some $K \geq 4$ on a connected strong $C_4$-free graph with $p \leq 2K$, or a connected strong $C_4$-free graph of radius $r \geq 5$ and induced-path order $2r - 1$ with $\ell > r - 1$. The graphs $F_2$ and $\theta_k$ of Corollary 6.3 have induced-path order 6 and $\ell = 3 - 5/n$ and $\ell = 3 - 6/n$ respectively, so there is no uniform gap below 3 from which a longer induced path would inherit $S \leq 3n$. The examples that follow limit extensions which do not use a new bound of $\mathrm{GEN}_K$ type.

**Conjecture 6.4.** *For every integer $n \geq 10$, the maximum of $S(G)$ over connected strong $C_4$-free $P_7$-free graphs of order $n$ equals $f(n)$ from Theorem 5.6. Every graph that attains the maximum is one of the extremal graphs of that theorem, and in particular is triangle-free.*

Conjecture 6.4 is open. It would unify Theorems 5.6 and 5.10: the triangle-free extremal graphs would be extremal in the whole class. The enumeration of the connected strong $C_4$-free $P_7$-free graphs recorded in the 2026 referee report, and implemented in `referee_checks/`, agrees with the conjecture for $10 \leq n \leq 20$. That computation is not a proof.

*Example 6.5.* Let $s \geq 3$, $t \geq 3$ and $m \geq 1$ be integers. Let $G_{s,t,m}$ be the cycle $C_{2s} = c_0 \cdots c_{2s-1}$ with a pendant path of $t$ edges at every cycle vertex, together with $m$ internally disjoint paths $c_0 x z y c_1$. Then $n = 2s(t + 1) + 3m$, $e = 2s(t + 1) + 4m$ and $T = 0$, so $e - n - T = m \geq 1$. For $(s, t, m) = (3, 3, 1)$ one has $n = 27$, $e = 28$, $r = 6$ and $p = 11$. These graphs satisfy REAL-133, since $\ell = 2e/n < 8/3$. The identity $e = n + T$ holds for every graph of Theorem 4.2. The count in the proof of Corollary 4.4 gives $n = 6 + a + b$, $e = 6 + a + 2b + m$ and $T = b + m$ in the construction, hence $e = n + T$, and $C_8$ has $e = n$ and $T = 0$. The graphs $G_{s,t,m}$ show that the identity fails for strong $C_4$-free graphs of induced-path order $2r - 1$ at every radius $r \geq 6$: here $r(G_{s,t,m}) = s + t$ and $p(G_{s,t,m}) = 2(s + t) - 1$.
Two vertices have at most one common neighbour, so $G$ is strong $C_4$-free and triangle-free. An added path of length four does not shorten distances among the older vertices, and $c_0$ remains within distance two of every vertex of such a path, so the eccentricity of $c_0$ is still $s + t$. A vertex $w$ of an added path meets the tip of the arm at $c_s$ only through $c_0$ or $c_1$, hence at distance at least $s + t$. Thus $r = s + t$. For the path order, let $B$ be the block formed by $C$ and the added paths. An induced path $P$ meets $B$ in a subpath $Q$ and can leave $B$ only at an end of $Q$ that is a cycle vertex, through at most $t$ arm

vertices. If $Q$ avoids both $c_0$ and $c_1$, then $|Q| \le 2s-2$. If $Q$ contains exactly one of them, then either $Q$ lies on the cycle with $|Q| \le 2s-1$, or $|Q| \le 2s+2$ with at most one cycle end, or $|Q| \le 7$ with no cycle end. If $Q$ contains both, the same three shapes occur with $|Q| \le 2s-1$, $|Q| \le 2s+2$ and $|Q| \le 6$. Adding at most $t$ arm vertices at each cycle end of $Q$ gives $|P| \le \max\{2s-1+2t,\ 2s+2+t,\ 7\}$. For $s \ge 3$ and $t \ge 3$ this maximum equals $2s+2t-1$, and the cycle minus one vertex together with the arms at its two ends attains that order.

*Example 6.6.* A uniform lower bound on the deletion loss does not bound $\lambda$. Two copies of the Petersen graph sharing a vertex form a triangle-free strong $C_4$-free graph on $n = 19$ vertices with $S = 60$, minimum deletion loss 6, induced-path order 9 and $\lambda = 6$ at the shared vertex. For $b \ge 2$, gluing $b$ copies of the Petersen graph along a common edge gives a 2-connected graph with $n = 8b+2$, $S = 28b+2$, minimum loss 6, induced-path order 9 and $\lambda = 2b+1$ at the two hubs.

*Example 6.7.* Let $q$ be a prime power, and let $F$ be the incidence graph of the projective plane $\mathrm{PG}(2,q)$, with $f = |F| = 2(q^2+q+1)$ and $S(F) = (q+1)f$. Choose a root of $F$, set $t = f-1$ and $b \ge 1$, and take a 6-cycle $c_0 \cdots c_5$. Attach a pendant path with $t$ edges at each cycle vertex, and attach $b$ copies of $F$ at $c_0$ by identifying their roots with $c_0$. Each new block meets the rest of the graph in a cut vertex, so the result is strong $C_4$-free, and

$$n = 6f + b(f-1), \qquad S = 12f + b(q+1)f, \qquad r = t+3, \qquad p = 2t+5 = 2r-1.$$

Distances inside the flower are unchanged. Every vertex of an attached copy is within distance $f-1 \le t$ of $c_0$, so each cycle vertex has eccentricity $t+3$, while a vertex $x$ of a copy has distance $d_F(x,c_0)+t+3$ from the tip of the arm at $c_3$. An induced path meets the cycle in at most five consecutive vertices, and each of its two ends outside the cycle contributes at most $t$ vertices. The sum $S$ is additive at cut vertices by Lemma 2.3. For fixed $q$, as $b \to \infty$ one has $\ell \to (q+1)f/(f-1)$. Letting $q \to \infty$, with $r = f+2$, this limit grows like $\sqrt{r/2}$. For $q = 5$ and $b = 17$ one has $n = 1409$, $r = 64$, $p = 127$ and $S = 7068 > 5n$. Thus $\ell$ is unbounded on the class of strong $C_4$-free graphs with $p = 2r-1$, and any statement $\mathrm{RAD}(r)$ has to use a bound that depends on $r$.

*Example 6.8.* Let $P$ be the Petersen graph, and let $u$ and $v$ be the ends of an induced $P_5$ in $P$. Such a path exists by the proof of Theorem 5.6, and $u$ and $v$ are nonadjacent. Form $G$ by adding one leaf at $u$ and one leaf at $v$. Then

$$\big(r(G), p(G), |V(G)|, S(G), \ell(G)\big) = (2, 7, 12, 34, 17/6).$$

The graph $P$ has 15 edges. The two new edges create no cycle, so $G$ is triangle-free and strong $C_4$-free, $n = 12$, $e = 17$ and $S = 2e = 34$. The graph $P$ is $P_6$-free by Theorem 3.6. A leaf has degree one, so it is an end of every induced path that contains it, and deleting the leaves that lie on an induced path of $G$ leaves an induced path of $P$. The path formed by the two leaves and the chosen $P_5$ is induced, because each leaf is adjacent only to its endpoint. Hence $p(G) = 7$. The diameter of $P$ is 2, so $u$ and $v$ have a common neighbour $w$. Every vertex of $P$ is at distance at most 2 from $w$, and each new leaf is at distance 2 from $w$. Thus the radius is 2. The example shows that $r = 2$ and $p = 7$ occur together with $\ell = 17/6$.

## APPENDIX

### A | The two-triangle certificate

Lemma 5.9 grows four labelled graphs, one from each way two triangles can meet. The authors' checker is `certificates/gen3/verify_gen3.py`, with SHA-256 digest `f1b67189 41f9cf3e 6b0917cf 25ab9093 810b07f6 d35a11fb 7cc539fb 546e6fd0`. The program uses only the Python standard library, requires Python 3.10 or later, embeds the decision certificate and takes no arguments. From the repository root the command is `python3 certificates/gen3/verify_gen3.py`. A run of that command replays the embedded certificate. It prints

```
triangle distance 0    nodes= 36  maximum order=17  leaves= 7  unrestricted=0
triangle distance 1    nodes= 31  maximum order=15  leaves= 6  unrestricted=0
triangle distance 2    nodes= 46  maximum order=11  leaves=21  unrestricted=0
triangle distance 3    nodes=  5  maximum order=10  leaves= 3  unrestricted=0
PASS. All branches and all decision entries verified.
```

The four rows are the searches of Lemma 5.9, in the order shared vertex, one joining edge, and joining paths of lengths two and three. Together they visit 118 nodes and 37 leaves. As in the proof of that lemma, a node is unrestricted when every vertex of the current graph has deletion loss at least 4. The unrestricted count on each of these four rows is zero, so every branch ends at a leaf.

The decision at each node of a distance row is the vertex of deletion loss less than 4 with the fewest legal children. Ties are broken by the smallest label, and the children of a node are visited in increasing order of the neighbourhood bitmask.

These two rules determine the arrays below. They are written in depth-first preorder, one integer per node, and the integer is the label of the chosen vertex. An independent re-implementation is `referee_checks/lemma57b.c`, from a JGT referee report (2026), reproduced in this repository for archival cross-verification and not written by the authors. With the environment variable `DEFONLY=1` it uses only deficient-vertex decisions and the same fail-first rule. On the seeds `D{c`, `E{CW`, `FwC[_` and `GwC[?c` it reproduces the four arrays and the counts above. Replaying `verify_gen3.py` is a check of the authors' certificate, not a second implementation. A Zenodo DOI for `referee_checks/` is pending.
The arrays, in the order of the four distance rows, are as follows.

```
0: 0, 1, 2, 1, 3, 4, 2, 3, 4, 6, 9, 9,
   6, 9, 9, 7, 7, 4, 10, 6, 6, 3, 9,
   6, 6, 2, 7, 9, 9, 1, 3, 4, 6, 8, 10, 10
1: 0, 3, 1, 2, 4, 5, 8, 8, 2, 1, 2, 1,
   2, 1, 3, 7, 1, 2, 4, 5, 10, 10, 0, 3,
   6, 1, 2, 4, 5, 10, 10
2: 0, 1, 2, 5, 4, 4, 5, 2, 5, 1, 5, 2,
   1, 2, 1, 1, 2, 5, 4, 7, 1, 2, 4, 2,
   1, 6, 6, 2, 6, 1, 6, 2, 1, 1, 2, 4,
   2, 1, 5, 5, 2, 1, 2, 5, 1, 5
3: 1, 6, 1, 6, 1
```

The nodes by order, and the orders of the leaves, are as follows. Shared vertex: 1, 1, 1, 1, 2, 2, 2, 3, 4, 5, 7, 5, 2 nodes on orders 5 through 17, and 2, 3, 2 leaves on orders 15, 16, 17. One edge: 1, 2, 3, 6, 6, 3, 3, 3, 2, 2 nodes on orders 6 through 15, and 3, 1, 2 leaves on orders 10, 13, 15. Path of length two: 1, 4, 14, 19, 8 nodes on orders 7 through 11, and 1, 12, 8 leaves on orders 9, 10, 11. Path of length three: 1, 3, 1 nodes on orders 8, 9, 10, and 2, 1 leaves on orders 9, 10. The distance-3 seed already spans an induced path on six vertices, by Lemma 5.8(4), and the search dies by order 10. The distance-2 search is the broadest, with 21 leaves, and it dies by order 11. The other two searches reach orders 17 and 15 and then close. No node in any of the four trees is free of a vertex of deletion loss less than 4. This is a summary of the closed branches, not a hand proof that each leaf is illegal.
At each node the verifier names the vertex just described and explores every legal neighbourhood that contains it. A child is rejected when two vertices of the new neighbourhood have a common neighbour in the graph already built. It is rejected when the new vertex and six existing vertices induce a $P_7$. The induced-$P_7$ test is the following. For a set $S$ of six vertices of the graph already built and a new vertex $z$, the set $S \cup \{z\}$ induces a $P_7$ precisely when $G[S]$ is a disjoint union of one or two paths and the neighbourhood of $z$ on $S$ consists of exactly one endpoint of each path component and of no other vertex of $S$. An isolated vertex is a path of order one and is its own end.

```
for each 6-set S:
    if some vertex of S has more than two neighbours in S: skip
    if the number of edges induced by S is not 4 or 5: skip
    split S into its components
    if a component has no end, or the number of components
        is not 6 minus the number of edges: skip
    reject each neighbourhood N for which N cap S consists
        of exactly one endpoint of each path component
        and of no other vertex of S
```

The reason each finite tree rules out a counterexample is the argument in the proof of Lemma 5.9.

| Initial configuration | Order | Nodes | Largest order | Leaves | Unrestricted |
|---|---|---|---|---|---|
| Shared vertex | 5 | 36 | 17 | 7 | 0 |
| One joining edge | 6 | 31 | 15 | 6 | 0 |
| Joining path of length two | 7 | 46 | 11 | 21 | 0 |
| Joining path of length three | 8 | 5 | 10 | 3 | 0 |

## DATA AVAILABILITY

The source of this article is https://github.com/neardws/induced-paths-without-4-cycles. The authors' certificate for Lemma 5.9 is `certificates/gen3/verify_gen3.py`. A re-implementation provided with a JGT referee report (2026), and reproduced for archival cross-verification, is the directory `referee_checks/`. A persistent DOI for that directory is pending. The map from claims of this article to scripts is in `referee_checks/README.md`. The directory `lean/` formalizes

the numbered lemmas and theorems of this article. Its numbered inventory has no remaining `sorry` or `admit`. Five literature axioms remain, recorded in `lean/AXIOMS.md`. That development is not a verification of this article from an empty axiom set.

## FORMALIZATION NOTE

The directory `lean/` is a Lean 4 and Mathlib formalization of the numbered lemmas and theorems of this article. The files are at

https://github.com/neardws/induced-paths-without-4-cycles/tree/main/lean.

The numbered inventory has no remaining `sorry` or `admit`. Five literature axioms remain, recorded in `lean/AXIOMS.md`. Finite certificates in that development are closed by `native_decide`, which trusts the Lean compiler and its runtime. This is not a verification of the article from an empty axiom set.


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